Project 4: Shooting star forecast

Documentation set, October 9, 2026

This PDF is built from the files in the Shooting Star Forecast folder. Links to other folders appear as plain text.

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Part 1: Overview

`Shooting Star Forecast/README.md`

Project 4: Shooting star forecast

Scope: Project 4 only (Shooting star forecast). The platform documents in `../Platform/` are shared by all five projects.

Documentation set for Project 4. Status: draft, awaiting John's review and approval. Last updated Oct 9, 2026.

Documents

Document	What it covers
requirements.md	Features, functional and quality requirements, acceptance criteria
design-doc.md	Architecture, components, data model, API, risks and open questions (section 16)
adrs/	Architecture decision records, 0001 to 0018
contracts/	ODCS data contracts for each source and published table
openapi.yaml	Draft API specification (GET /forecast, POST /votes in Phase B)
operations.md	Schedule, freshness, monitoring, what happens when something fails, runbooks, cost
security-and-privacy.md	Threat model, personal-data inventory, deletion periods, licence and credit register
changelog.md	Record of each revision

Key points

- The sky score is a documented formula, not a model (ADR-0001). The formula and starting weights are in ADR-0010, approved by John on Oct 9, 2026.
- Phase A covers the lower 48 states. Alaska and Hawaii follow the VIIRS check (ADR-0006).
- A missing score for tonight shows the latest earlier data with a late notice (ADR-0011).
- Source licences are checked against each provider's page (ADR-0014). Some items are unverified.

Ready for John's approval

Project 4's six document types are drafted and complete. Coding of Phase A (4A) starts only after John approves them and the platform set [Documentation Plan: How the documents get drafted].

Documents to approve

Document	Status
requirements.md	Draft, complete; open items listed below
design-doc.md	Draft, complete; section 16 holds the open questions
adrs/ (0001 to 0018)	0010 Accepted (formula); 0014 Accepted (licences); 0017 Default; 0018 Accepted; the rest Accepted
contracts/ and openapi.yaml	Draft; YAML parses; the score fields follow ADR-0010
operations.md	Draft, written from the design doc, ADRs and contracts; undecided values name their open question
security-and-privacy.md	Draft, written from the same sources; confirms "no personal data intended" only once OQ-10 and OQ-15 are answered
Platform set (../Platform/)	Approved by John, Oct 9, 2026 (OQ-P01, OQ-P02). The idle-alarm duration is still to be measured.

Decisions still needed from John before 4A.2 starts

- OQ-13: the numeric margin for the "clearly different" check (ADR-0015).

- OQ-03: the time zone source and the sunset and sunrise method.
- OQ-06: score grain, cloud-to-cell assignment, lower-48 boundary source, and the land-cell definition.
- OQ-14: whether the IMO calendar gives the active dates and what the rate means.
- OQ-20: the freshness threshold for the publish gate, and retry counts and timeouts.
- OQ-18: confirm the credit lines and the unverified items in ADR-0014.

Spikes that gate Phase A (each needs running something; see design doc section 16)

- OQ-01: VIIRS latitude coverage. OQ-04: GFS run, hours and format. OQ-05: file sizes and job limits. OQ-07: VIIRS product details (download is by hand, ADR-0018). OQ-24: which annual VIIRS products exist.

Platform gates (the platform set is in ../Platform/)

- Platform set approved (OQ-P01, OQ-P02, Oct 9, 2026). The idle-alarm duration is still to be measured (OQ-P02).
- Walking skeleton steps 0.1 to 0.5 in place before 4A coding; the Phase A gate in requirements.md section 5.

Files in this folder

Everything for Project 4 is in this folder: the documents, the ADRs, the contracts, the API spec, and the PDF `full-documentation.pdf`. Two things are shared and stay outside:

- `..\Platform\`: the platform documents, shared by all five projects.

File or folder	What it is
README.md	This index and the approval checklist
requirements.md	Requirements, quality scenarios, acceptance criteria
design-doc.md	Design doc (17 sections), including open questions (section 16)
decisions.md	Project 4's answered decisions and open items (split from the shared logs, Oct 9, 2026)
changelog.md	Record of each revision
openapi.yaml	API spec (GET /forecast; POST /votes in Phase B)
operations.md	How the project runs day to day: schedule, freshness, monitoring, failures, runbooks, cost
security-and-privacy.md	Threat model, personal data, deletion periods, licence and credit register
contracts/	Seven data contracts: one per source or published table
adrs/	Decision records 0001 to 0018
full-documentation.pdf	All of this project's documents in one PDF, rebuilt for revision 8

PDF

`full-documentation.pdf` is built from this folder and is current as of revision 8 (Oct 9, 2026). It contains the overview, requirements, design doc, operations, security and privacy, ADRs, data contracts, API spec, changelog, and the raw YAML of the contracts and API.

Earlier PDF revisions (the original and revisions 2 to 6) were removed as old versions. Their history is in changelog.md.

Project 4 decisions

Project 4's decisions are in decisions.md: the answers John gave on Oct 9, 2026, and the open items from the first review. The original shared logs are kept, unchanged, in `..\_archive\`.

Open questions

Section 16 of the design doc holds the open questions (OQ-01 to OQ-24). Answered questions stay in the table with the answer and the ADR number.

Part 2: Requirements

`Shooting Star Forecast/requirements.md`

Project 4: Shooting star forecast. Requirements

Field	Value
Status	Draft, awaiting John's review and approval
Owner	John (GitHub: jwasikye)
Reviewers	John approves each document before coding of this part begins [DP: How the documents get drafted]
Last updated	Oct 9, 2026
Companion documents	design-doc.md, adrs/, contracts/, openapi.yaml
Platform documents	../Platform/design-doc.md (draft, Oct 9, 2026, awaiting John's approval)

Source key

Every value in this document is a recorded decision or a confirmed fact. Citations:

- **[BP x]**: Build Plans for the Five Projects, step or section x (for example BP 4A.1).
- **[BD x]**: Build Plan Decisions, the row named x.
- **[DP x]**: Documentation Plan, section x (including "Answered by research").
- **[ADR-00nn]**: an architecture decision record in adrs/. "Decided by John, Oct 9, 2026" marks a decision John made; ADR-0010 (score formula) was approved by John on Oct 9, 2026, as written.
- **[OQ-nn]**: an open question in design-doc.md section 16. Anything not determined by the sources is an open question and is not guessed here.

Names of tables, columns, endpoints, functions and files in this document set are proposed by the design documents. They become binding when John approves the documents [DP: How the documents get drafted].

Terms used throughout (the full glossary is in design-doc.md section 17):

- **Phase A, B, C**: the build plan builds each project in three phases. Phase A is the smallest version that works end to end and goes live. Phase B adds depth. Phase C adds scale and is built in a second round after every project reaches Phase B.
- **Session box**: the temporary AWS computer (a spot instance, a cheaper computer that AWS can take back at short notice) that runs Airflow and Spark. It is switched off most of the day.
- **Nightly run**: the scheduled pipeline that starts the session box at 10:00 UTC, processes all inputs, publishes the results and shuts the box down.
- **Cell**: one hexagon of the H3 grid. H3 is a system that divides the Earth into hexagons of a chosen size (called a resolution).
- **Top factor**: the one factor (cloud cover, Moon, light pollution or no active shower) that takes the most points off the score. The name "no active shower" was approved by John on Oct 9, 2026, with the formula (OQ-02).
- **Late notice**: the visible message the page shows next to old data when tonight's data has not arrived, together with the time of the last good run.
- **CI**: the automatic tests that GitHub runs on every change.

- **Engineering standards** cited in this document are numbered rules from the build plan's Engineering standards table: 1 keep raw files unchanged with source URL, fetch time and hash (public archives that never change get a manifest of URL, size and hash instead); 2 carry event time and a source key on every record; 3 make tasks idempotent and atomic (overwrite a partition, never append blindly); 4 write, audit, publish (new data lands in staging, passes checks, then goes live); 5 give each source a data contract and diff the schema before loading; 6 layer dbt as staging, intermediate and marts; 7 run Iceberg maintenance as the last task of the nightly run; 9 pick the H3 resolution from the source and join with cell_to_parent; 11 make pipelines observable (freshness and row counts per run, a status page, one ticket per incident); 13 least privilege and every key in Parameter Store; 15 store every timestamp in UTC; 16 anything that runs around the clock writes only to always-on storage.
- **Tools** named in this document, in plain words: Kafka (a message queue for live data; this project does not use it); Snowflake (a cloud data warehouse; not used here); MetricFlow (a tool that defines business metrics; not used here); Superset (a dashboard tool; not used here); Grafana (a dashboard tool for system health; not used here); OpenLineage (a standard for records of which data each job read and wrote); Terraform (a tool that creates cloud resources from code); OIDC (OpenID Connect, which lets GitHub Actions get short-lived AWS access without stored keys); FastAPI (a Python web framework); API Gateway (the AWS service that exposes the Lambda function over HTTP); CloudFront (the AWS service that serves the site's files worldwide); X-Ray (the AWS service that records traces of requests); Fargate (AWS service that runs scheduled containers); CloudWatch (AWS monitoring); Object Lock (an S3 setting that blocks deletion or overwrite of a file for a set time); SCD Type 2 (slowly changing dimension, a way to keep old versions of changing rows); LLM judge (a language model that scores another model's answers; not used here).

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1 Introduction and goals (arc42 section 1)

1.1 Purpose

The project answers one question: tonight's chance of seeing a shooting star at a location, and the top factor behind that chance. The chance is a score computed by a documented formula from four inputs: cloud cover, the Moon, light pollution (measured by VIIRS, a satellite that records nighttime lights) and the meteor shower calendar. The project uses no AI model on purpose, and a decision record (ADR-0001) says why a model would not improve the score [BP Project 4].

1.2 Users and needs

The sources define no user personas beyond the site's visitors and the owner. The needs below are derived from the features in BP Project 4 and are limited to what those features support.

User	Need	Served by
Visitor (anyone looking at the public site)	Enter a place and see tonight's score and the top factor behind it	F-A3
Visitor	See where each number comes from, with every source credited	F-A1, F-A3
Visitor	Know when the data is late, with the time of the last good run, instead of seeing nothing	F-A3
Visitor	Get an answer for a place outside the United States, and be told where the data stops (polar limits)	F-B1
Visitor	Read a plain phrase that explains the top factor with the actual values (for example, the Moon is 92% lit and drowns out the shower)	F-B2
Visitor	Report whether they saw a shooting star ("did you see one?")	F-B3
Visitor (second round)	Hourly forecast updates; an optional email the day before a good night at a saved location	F-C1, F-C2
Owner (John)	A page that works with the session box off, costs nothing surprising, and shows honest limits	QR-01, QR-06
Owner (John)	A score that can be audited: documented weights, named top factor, checkable inputs	F-A2, QR-04, QR-05

1.3 Quality goals

The build plan does not rank quality goals. They are listed without priority:

1. The lookup answers with the session box off [BP 4A.3].
2. The score is explainable: a documented formula, a named top factor, phrases whose values match the table [BP 4A.2, 4B.2; ADR-0001, ADR-0002].
3. Inputs are trustworthy and credited: each source has a data contract (a written description of its columns and rules), a publish gate (an automatic check that blocks bad data), a terms note and a credit line [BP 4A.1; the same approach as Project 1 step 1A.5].
4. Nothing surprises on cost: hard limits on every paid item [BP Hard limits table].
5. Data is fresh every day: the nightly run starts at 10:00 UTC [BD Nightly run time; BD Data freshness].

1.4 Scope boundary

In scope: the nightly pipeline that produces tonight's score per cell, the published tables, the lookup endpoint, the lookup page, and in Phase B the explanation phrases, the backtest and the "did you see one?" votes.

Out of scope for this document set:

- Phase C (hourly forecast updates, optional email). Phase C starts only after every project reaches Phase B [BP How the plan works]. It is listed as features F-C1 and F-C2 with no acceptance criteria because the plan records none. Decided by John, Oct 9, 2026: Phase C stays unspecified and needs its own design (hourly updates conflict with the model where the session box is off most of the day); its done-when checks are missing and are written when Phase C is designed.
- Platform content (CI, Terraform, session box, serving template, status page, incident record). Linked, not repeated.
- The "while building" documents: README, reference from code, operations, security and privacy, changelog. No model card, because there is no model [DP: Which documents each part gets].

1.5 Constraints

- Budget: under $50 a month for the whole plan, browser only [BD Budget]. This project adds no model API spend because it makes no model calls (see ADR-0001).
- Every paid item has a hard limit, including maximum rows and runtime per Spark job [BD Surprise costs; BP Hard limits table].
- Portfolio, non-commercial: licensing must be checked before charging for any project [BP Assumptions].
- Timestamps stored in UTC, converted only on the page [BD Timestamps; BP Engineering standard 15].
- The dev copy (for trying things) and the live copy (the public site) are fully separate. Only merges to the main branch reach live, and only after the changed pipelines pass tests on sample data in CI [BP 0.2].
- Source terms: NOAA asks for attribution for GFS [DP Cloud cover data]; the VIIRS annual composite is not confirmed as public domain. EOG's own page says many VIIRS nighttime lights products are CC BY 4.0, and the annual composite's licence is unverified [ADR-0014; OQ-18]. The IMO (International Meteor Organization) site has a copyright notice and no reuse terms, so the project stores only factual fields and credits the IMO [BP Meteor shower calendar; ADR-0005].

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2 Features by phase

ID	Phase	Feature	Plan step	Technology (from the Technology by project table)
F-A1	A	Inputs: GFS cloud cover (NOAA's Global Forecast System), VIIRS annual nighttime lights, Moon position (computed with the Skyfield library), IMO shower calendar	4A.1	Spark, Airflow, Iceberg, Glue, H3
F-A2	A	Score: per-cell, per-night conditions, the score, the top factor and documented weights	4A.2	Spark (PySpark), dbt (DuckDB), Great Expectations

ID	Phase	Feature	Plan step	Technology (from the Technology by project table)
F-A3	A	Lookup page: enter a place, see tonight's score and top factor, sources credited	4A.3	React, FastAPI, API Gateway, DuckDB serving in Lambda
F-B1	B	Global coverage: every land cell the VIIRS data covers, latitude limits stated on the page	4B.1	same as Phase A
F-B2	B	Explanations: pre-written phrases for each top factor, filled with the computed values	4B.2	same as Phase A
F-B3	B	Backtest and feedback: compare scores with known shower peaks and with Project 2's fireball baseline (its record of observed fireballs), check 20 Moon positions against JPL Horizons (NASA's online positions service), "did you see one?" button with votes in DynamoDB	4B.3	DynamoDB, OpenLineage and dbt docs (Phase B)
F-C1	C	Hourly forecast updates	Phase C	not specified in the plan
F-C2	C	Optional email the day before a good night at a saved location	Phase C	not specified in the plan

The build plan's Technology by project table marks these tools for Project 4: Spark, Airflow, Iceberg with Glue, dbt, Great Expectations, DuckDB serving, H3, React, FastAPI, API Gateway, Terraform, Docker, k3s and CI from Phase A; DynamoDB, OpenLineage and the dbt docs catalog from Phase B. The table marks none of Kafka, Snowflake, MetricFlow, Superset, Grafana, Databricks, scikit-learn, MCP servers, agents, evals or LLM judge for Project 4. Full names are in the design doc glossary.

First-run checks (assumption tests)

Phase	Check	Source
A	GFS cloud files read correctly from AWS	BP Project 4 Phase A
A	The VIIRS download works with a free Earth Observation Group account (see OQ-23)	BP Project 4 Phase A
B	20 Moon positions from Skyfield match JPL Horizons within a stated tolerance. This is the same test as the 4B.3 acceptance criterion (FR-37), which is written once, under F-B3	BP Project 4 Phase B

The build plan records no fallback for these checks (what to do if a check fails), and none is decided. The related open questions are [OQ-04], [OQ-07] and [OQ-09].

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3 Functional requirements

Phase A (live). Scope: the lower 48 states first. Alaska and Hawaii are added once the VIIRS first-run check shows which latitudes the VIIRS file covers (decided by John, Oct 9, 2026; ADR-0006).

ID	Requirement	Feature	Source
FR-01	The component gfs_ingest (design doc 7.2) shall store NOAA GFS total cloud cover (percent, whole atmosphere, 0.25 degree grid) from the AWS open data bucket noaa-gfs-bdp-pds (region us-east-1) as H3 resolution 4 cells, writing the raw file and the table rows, and shall record the load time.	F-A1	BP 4A.1; DP Cloud cover data
FR-02	The component viirs_ingest (design doc 7.3) shall store the VIIRS annual nighttime lights composite as H3 resolution 8 cells and record the load time. It shall keep a manifest (URL, size and hash, where a hash is a fingerprint of the file contents) instead of the raw file, and the yearly load shall read the file again once a year (John downloads it by hand, ADR-0018) (decided by John, Oct 9, 2026).	F-A1	BP 4A.1; ADR-0012
FR-03	The component moon_calculator (design doc 7.5) shall compute Moon altitude and Moon illumination with the Skyfield library and a JPL ephemeris (a data file of planetary and lunar positions).	F-A1	BP 4A.1; ADR-0003

ID	Requirement	Feature	Source
FR-04	The component imo_calendar (design doc 7.4) shall store the International Meteor Organization (IMO) shower calendar as a small reference table that contains only the fields the score needs, each with its source: shower name, active dates, peak date and rate. Radiant position (the point in the sky a shower appears to come from) is stored only if the formula uses it. Decided by John, Oct 9, 2026.	F-A1	BP 4A.1, Meteor shower calendar; ADR-0005
FR-05	Each input component (gfs_ingest, viirs_ingest, imo_calendar and moon_calculator) shall attach a terms note (what the source allows) and a credit line (the text that credits the source) to its input.	F-A1	BP 4A.1
FR-06	Each input component shall keep raw files unchanged with the source URL, fetch time and hash, and shall put event time and a source key on every record. For public archives that never change it keeps only a manifest of URL, size and hash; VIIRS is handled this way (decided by John, Oct 9, 2026).	F-A1	BP Engineering standards 1, 2; ADR-0012
FR-07	The PySpark job build_conditions (design doc 7.6) shall build per-cell, per-night conditions (cloud cover, Moon, light pollution, shower activity), and the score engine (design doc 7.7) shall compute the score. A night is the local night at the cell, from sunset to sunrise. The job evaluates the Moon across the whole night and stores times in UTC (decided by John, Oct 9, 2026).	F-A2	BP 4A.2; ADR-0013
FR-08	The score job shall name the top factor for each score.	F-A2	BP 4A.2
FR-09	The project shall document the score weights. ADR-0010 sets the formula and its starting weights, approved by John on Oct 9, 2026; the weights are not validated, and the 4B.3 backtest tunes them later.	F-A2	BP 4A.2; ADR-0010
FR-10	The dbt project (dbt builds data models from SQL) shall run on DuckDB and be layered as staging, intermediate and marts (final tables). Only the staging layer shall read raw sources (source()), and unique and not_null tests shall cover the keys.	F-A2	BP 4A.2, Engineering standard 6
FR-11	The publish gate (a Great Expectations check) shall test forecast freshness and value ranges, and the publisher shall not publish when a check fails (write, audit, publish: new data lands in staging, passes checks, then goes live).	F-A2	BP 4A.2, Engineering standard 4
FR-12	The nightly pipeline shall publish the final (gold) tables as Parquet files, which the lookup endpoint reads.	F-A3	BP 0.4
FR-13	The lookup page shall let a visitor enter a place and shall show tonight's score and top factor, with every source credited.	F-A3	BP 4A.3
FR-14	The lookup page shall reuse Project 1's map and the shared serving template.	F-A3	BP 4A.3; ADR-0008
FR-15	The lookup endpoint shall run as FastAPI on AWS Lambda behind API Gateway (HTTP type) and shall use DuckDB to read the published Parquet files, so it answers with the session box off.	F-A3	BP 0.4, Tools table; BD Serving queries
FR-16	The page shall state: "The first request after a period of inactivity may take up to 10 seconds while the service initializes."	F-A3	BP 0.4
FR-17	The nightly run (10:00 UTC, spot instance only) shall compute tonight's score from that morning's GFS forecast.	F-A2	BD Nightly run time; ADR-0007
FR-18	Each pipeline task shall be idempotent (safe to run again with the same result) and atomic (all of its output or none). It shall overwrite a partition (one slice of a table, here one night) instead of appending blindly, and shall use the run's data interval instead of the current time.	F-A1, F-A2	BP Engineering standard 3
FR-19	Each source shall have a data contract (schema, rules, freshness target, owner, version), and the pipeline shall compare the incoming schema with the contract before loading.	F-A1	BP Engineering standard 5; DP Interface specs
FR-20	Every Airflow DAG (a scheduled pipeline definition) shall write freshness and row counts to the metrics table. The status page shall show the project's pipelines. The platform shall send an email when a freshness target is missed or a run fails.	F-A2	BP 0.5, Engineering standard 11
FR-21	Every Spark job shall have a maximum number of rows read and written and a maximum runtime. A job over either limit stops before publishing and opens an incident. John sets the values from the measured size estimate ([OQ-05]).	F-A2	BP Hard limits table

ID	Requirement	Feature	Source
FR-22	The pipeline shall store all timestamps in UTC, and the page shall convert them to local time.	F-A2, F-A3	BD Timestamps; BP Engineering standard 15
FR-23	When tonight's score is late or missing, the page shall show the old data with a visible late notice and the time of the last good run. It shall not hide the data. Decided by John, Oct 9, 2026, for all five projects.	F-A3	ADR-0011

Phase B (depth).

ID	Requirement	Feature	Source
FR-30	The pipeline shall cover every land cell the VIIRS data covers, and the page shall state the latitude limits of that data.	F-B1	BP 4B.1
FR-31	The lookup page and endpoint shall return a score for places outside the United States.	F-B1	BP 4B.1
FR-32	For places at latitudes the VIIRS data does not cover (the polar limits), the page shall show the stated limit instead of a score.	F-B1	BP 4B.1
FR-33	The repository shall hold pre-written phrases for each top factor and for each combination of factors the formula can produce, and the page shall fill them with the computed values. The set of combinations is fixed once John approves the formula (OQ-02). No text is generated by a model.	F-B2	BP 4B.2; ADR-0002
FR-34	Every value shown inside a phrase shall equal the corresponding value in the score table.	F-B2	BP 4B.2
FR-35	The backtest job shall compare scores with known shower peaks and with the fireball baseline from Project 2.	F-B3	BP 4B.3
FR-36	The backtest job shall write a short note that shows where the score agrees and disagrees with known peaks.	F-B3	BP 4B.3
FR-37	An automated test shall compare 20 Moon positions from Skyfield with JPL Horizons and pass when each agrees within a stated tolerance.	F-B3	BP 4B.3, Project 4 Phase B first-run check
FR-38	The page shall have a "did you see one?" button, and the vote endpoint shall store each vote in DynamoDB (AWS's always-on database).	F-B3	BP 4B.3; BD DynamoDB; ADR-0009
FR-39	The system shall count the stored votes.	F-B3	BP 4B.3
FR-40	The site shall emit usage events that contain no raw IP address, in the format defined by the portfolio analytics of Project 2 step 2B.4.	F-B3	BP 2B.4 (covers all five sites)
FR-41	The project's pipelines shall emit OpenLineage events (records of which data each job read and wrote) and shall appear in the dbt docs catalog (a generated website describing the models).	F-B1, F-B2, F-B3	BP Technology table; BP 1B.3
FR-42	The vote endpoint and the page shall follow Project 1's privacy rules (decided by John, Oct 9, 2026): the phone creates an anonymous random ID and keeps it; limits apply per visitor and per day; no internet addresses are stored; the phone computes the H3 cell; and the limit counters are stored apart from the votes. The limit numbers are open ([OQ-15]).	F-B3	ADR-0009

Explicit non-requirements: the system makes no model API calls and has no model card, no eval harness (a set of tests that score model answers), no agent and no Snowflake marts. The Technology table marks none of these for Project 4; [OQ-12] asks John to confirm the Snowflake point.

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4 Quality requirements (arc42 section 10)

4.1 Quality tree

- Availability: QR-01
- Performance: QR-02

- Freshness: QR-03
- Correctness and explainability: QR-04, QR-05
- Cost: QR-06
- Privacy: QR-07
- Licensing and credit: QR-08
- Reliability and repeatability: QR-09, QR-10, QR-11
- Security: QR-12
- Observability: QR-13
- Change detection: QR-14

4.2 Quality scenarios

Where the sources record no number, the number is an open question and the row names it. No target is guessed.

ID	Quality	Scenario and target	How verified	Source
QR-01	Availability	The session box is off. A visitor looks up a place on the public link and receives a score and top factor.	With the session box stopped or destroyed, call the public link for a sample of places. Whether the deploy pipeline runs this check automatically is open ([OQ-19]).	BP 4A.3; BD Always-on parts
QR-02	Performance	The page shows the cold-start notice (the wording quoted in FR-16, which is the build plan's wording), and the project measures and reports the median warm latency and the 95th-percentile cold-start time of the Lambda function. No separate Project 4 latency target is recorded.	Check that the notice appears on the page, and report the measured times by applying the cold-start measurement of platform step 1B.9 to this endpoint. Project 4 targets are open ([OQ-19]).	BP 0.4, 1B.9; BD Cold starts
QR-03	Freshness	The pipeline runs every night at 10:00 UTC. About 45 minutes is expected and 2 hours is the hard maximum; reaching the maximum is recorded as an incident. The forecast-freshness threshold the publish gate uses is not recorded ([OQ-20]).	Read the per-run freshness and row counts in the metrics table, check the status page for the last good run, and test the publish gate's freshness check with sample files in CI.	BD Nightly run time, Nightly shutdown; BP 0.5, 4A.2
QR-04	Correctness (Moon)	Skyfield with a JPL ephemeris agrees with the U.S. Naval Observatory to within 0.00001 arcseconds (confirmed fact). The 20 Moon positions agree with JPL Horizons within a stated tolerance, which is open ([OQ-09]).	An automated test compares the 20 positions with Horizons reference values and runs in CI.	DP Moon calculation accuracy; BP 4B.3
QR-05	Explainability	The score comes from a documented formula, each score names its top factor, and each explanation phrase contains only values that equal the score table.	Unit tests of the score functions; in Phase B, an automated comparison of every phrase value with the score table.	BP 4A.2, 4B.2
QR-06	Cost	Every Spark job stops before publishing when it exceeds its maximum rows or runtime, and opens an incident. The nightly run has a 2-hour maximum. DynamoDB capacity is provisioned within the free tier, so heavy traffic is throttled instead of billed. The project makes no model API calls. The overall AWS budget guard is $40.	A CI test gives a job a limit below its input size and checks that the job stops and publishes nothing. Terraform shows the provisioned DynamoDB capacity. The budget action is a platform check. The limit values are open ([OQ-05]).	BP Hard limits table; BD Cost guards
QR-07	Privacy	Timestamps are in UTC. Usage events from the site contain no raw IP address. Votes follow Project 1's rules (anonymous phone ID, limits per visitor and per day, no internet addresses; decided by John, Oct 9, 2026); the numbers are open ([OQ-15]). Privacy of lookup requests is open ([OQ-10]).	Review the usage-event and vote schemas, and test that no IP field and no visitor ID reaches the votes table.	BP 2B.4; ADR-0009

ID	Quality	Scenario and target	How verified	Source
QR-08	Licensing and credit	Each of the four inputs has a terms note and a credit line, and the lookup page shows them. The credit for the Moon calculation follows the JPL and Skyfield terms ([OQ-18]).	CI checks the input tables and the source_credits table; a page check confirms that every credit line appears for a looked-up place.	BP 4A.1, 4A.3; DP Cloud cover data
QR-09	Repeatability	Running a night again overwrites only that night's partition, and the row count stays the same.	A CI test runs the job twice on sample data and compares counts and a content hash.	BP Engineering standard 3
QR-10	Data quality	Bad data never reaches the site, because publishing stops when a check fails.	A CI test feeds a sample file that violates a freshness or range check and confirms the publishing step is not reached.	BP Engineering standard 4; 4A.2
QR-11	Reliability on spot	Every job is safe to run again, because AWS can take a spot instance back with a two-minute warning. The nightly run uses spot instances only; when none is available, the night is skipped. A skipped night cannot be caught up, so the page shows the latest earlier data with a late notice and the time of the last good run (decided by John, Oct 9, 2026).	Run the rerun test (QR-09) and a late-data test with a fixture; the spot interruption handler is a platform check.	BP How the box is run; ADR-0011
QR-12	Security	Each pipeline has one least-privilege IAM role (AWS permissions limited to what it needs). Dev roles cannot write live data. Every key is in Parameter Store (AWS's encrypted key storage), none in code or GitHub. Container images are scanned.	Platform checks (a dev role is refused when writing live data); gitleaks (secret scanner) and Trivy (image scanner) run in CI.	BP 0.1, 0.2, Engineering standard 13
QR-13	Observability	The platform records freshness and row counts per run, emails on a failed run or missed freshness target, keeps one incident record per real incident, and opens one de-duplicated GitHub issue per failure (from platform step 1B.9).	The status page shows the last good run; the first real failure produces an email and a record.	BP 0.5, 1B.9
QR-14	Change detection	A source format change is caught before the nightly run fails. The pipeline compares the schema with the contract before loading, and a weekly smoke test pulls a small sample from every source the project uses. How the weekly test samples VIIRS, which needs a login, is open ([OQ-07]).	A contract-diff test in CI, and the weekly smoke test result.	BP Engineering standard 5, 1B.9; BD Source changes

5 Acceptance criteria

The criteria are the build plan's "Done when" checks. Where the plan states a check loosely, the verification column says how to make it testable and names the open question that settles it.

Feature	Criterion (build plan "Done when")	Verification	Prerequisites
F-A1 (4A.1)	All four inputs land with load time, each with a terms note and credit line. ("Land" is the build plan's word for stored, meaning the raw file and the table rows are written.)	After a run, query the input tables (where each input is first stored). GFS, VIIRS, Moon and IMO calendar each have rows with a non-null load time (UTC), and the source_credits table has four rows with a non-empty terms_note and credit_line. Contract checks pass in CI.	Both Phase A first-run checks passed
F-A2 (4A.2)	A full-Moon night and a night near a shower peak give clearly different scores and top factors.	Score two fixed nights from sample data, one with Moon illumination near full and one near a shower peak, and assert that the score and top factor differ. Until John approves the formula the check is written as in the plan; a numeric margin is set after approval (decided by John, Oct 9, 2026; [OQ-13]).	F-A1; ADR-0010 approved ([OQ-02])

Feature	Criterion (build plan "Done when")	Verification	Prerequisites
F-A3 (4A.3)	The public link answers a lookup with the session box off.	With the session box stopped, enter a place on the public link and check that tonight's score, top factor and all four source credits appear. Added by John's late-data decision: with tonight's row absent in a test fixture, the page shows the latest earlier row with the late notice and the time of the last good run.	F-A2 published; Project 1 map and serving template
F-B1 (4B.1)	A lookup outside the United States works, and polar limits are shown.	Look up a place outside the United States and a place beyond the VIIRS latitude coverage; the first returns a score, the second shows the stated limit. Limits come from [OQ-01].	OQ-01 answered
F-B2 (4B.2)	Every phrase's values match the score table.	An automated test renders every phrase for every score row (or for a sample if the full table is too large) and compares each embedded value with the table.	F-B1; phrase set ([OQ-02])
F-B3 (4B.3)	A short note shows where the score agrees and disagrees with known peaks, and votes are counted.	The note exists and cites its comparison against IMO peak dates and the Project 2 fireball baseline. The 20 Moon positions pass the Horizons test. Test votes are submitted and the stored count is checked. Where the note is published and where counts are shown are open ([OQ-15], [OQ-16]).	F-A2; Project 2 fireball data; DynamoDB table
F-C1, F-C2	No criteria are recorded in the plan. Decided by John, Oct 9, 2026: Phase C stays unspecified, needs its own design (it conflicts with the model where the session box is off most of the day), and its done-when checks are written when Phase C is designed.	not applicable	Phase B shipped

First-run check acceptance (per phase):

- Phase A: GFS cloud files read correctly from AWS; VIIRS download works with a free Earth Observation Group account (see OQ-23).
- Phase B: the 20-position Moon check is the same test as the F-B3 criterion above and is not a separate acceptance item.

Gate for Project 4 (the condition for moving on): after Phase A, tonight's score is public for the lower 48 states, then for Alaska and Hawaii. After Phase B, global coverage and the backtest are done [BP Build order and checkpoints; ADR-0006].

---

6 Traceability

Plan step	Features	Requirements	Quality
4A.1	F-A1	FR-01 to FR-06, FR-18, FR-19	QR-08, QR-09, QR-14
4A.2	F-A2	FR-07 to FR-11, FR-17, FR-20 to FR-22	QR-03 to QR-06, QR-10, QR-13
4A.3	F-A3	FR-12 to FR-16, FR-23	QR-01, QR-02, QR-08
4B.1	F-B1	FR-30 to FR-32	QR-08
4B.2	F-B2	FR-33, FR-34	QR-05
4B.3	F-B3	FR-35 to FR-42	QR-04, QR-07

7 Decisions needed from John before 4A.2 starts, and other blockers

Approval and decisions still needed from John (these are choices, not tests that need running): OQ-02 (approved Oct 9, 2026, ADR-0010), OQ-06 (score grain, lower 48 boundary source, land cell), OQ-03 (time zone source and sunset method), OQ-20 (freshness threshold), OQ-13 (the numeric margin, after OQ-02). Spikes (short tests that answer a question by running something) that gate Phase A: OQ-01, OQ-04, OQ-05, OQ-07, OQ-14. Added Oct 9, 2026 from

the source checks: OQ-23 (EOG programmatic access is paid-only from June 1, 2026; answered: the VIIRS file is downloaded by hand, ADR-0018), OQ-24 (no newer annual VIIRS product is listed) and OQ-22 (which night is "tonight" before sunrise).

Other items that depend on open questions

The full list with GitHub issue placeholders is in design-doc.md section 16, with the questions John answered on Oct 9, 2026 listed there. The ones that block a requirement from being testable: OQ-01 (VIIRS latitudes), OQ-02 (formula, approved Oct 9, 2026), OQ-05 (data sizes and job limits), OQ-09 (Moon test tolerance), OQ-13 (numeric margin), OQ-19 and OQ-20 (latency and freshness numbers).

Part 3: Design document

Shooting Star Forecast/design-doc.md

Project 4: Shooting star forecast. Design doc

Template: Google design doc, arc42 sections, C4 model [DP: Design doc template]. Citations: [BP x] means Build Plans for the Five Projects, step or section x; [BD x] means Build Plan Decisions; [DP x] means Documentation Plan; [ADR-00nn] means a decision record (architecture decision record) in adrs/; [OQ-nn] means an open question in section 16. "Decided by John, Oct 9, 2026" marks a decision John made; ADR-0010 (the score formula) is only Proposed. Phase A, B and C are the build plan's three phases: A is the smallest live version, B adds depth, C adds scale in a second round. Engineering standards cited in this document are numbered rules from the build plan's Engineering standards table: 1 keep raw files unchanged with source URL, fetch time and hash (public archives that never change get a manifest of URL, size and hash instead); 2 carry event time and a source key on every record; 3 make tasks idempotent and atomic (overwrite a partition, never append blindly); 4 write, audit, publish (new data lands in staging, passes checks, then goes live); 5 give each source a data contract and diff the schema before loading; 6 layer dbt as staging, intermediate and marts; 7 run Iceberg maintenance as the last task of the nightly run; 9 pick the H3 resolution from the source and join with cell_to_parent; 11 make pipelines observable (freshness and row counts per run, a status page, one ticket per incident); 13 least privilege and every key in Parameter Store; 15 store every timestamp in UTC; 16 anything that runs around the clock writes only to always-on storage. Tools named in this document, in plain words: Kafka (a message queue for live data; this project does not use it); Snowflake (a cloud data warehouse; not used here); MetricFlow (a tool that defines business metrics; not used here); Superset (a dashboard tool; not used here); Grafana (a dashboard tool for system health; not used here); OpenLineage (a standard for records of which data each job read and wrote); Terraform (a tool that creates cloud resources from code); OIDC (OpenID Connect, which lets GitHub Actions get short-lived AWS access without stored keys); FastAPI (a Python web framework); API Gateway (the AWS service that exposes the Lambda function over HTTP); CloudFront (the AWS service that serves the site's files worldwide); X-Ray (the AWS service that records traces of requests); Fargate (AWS service that runs scheduled containers); CloudWatch (AWS monitoring); Object Lock (an S3 setting that blocks deletion or overwrite of a file for a set time); SCD Type 2 (slowly changing dimension, a way to keep old versions of changing rows); LLM judge (a language model that scores another model's answers; not used here). Names of tables, columns, endpoints, functions, classes and files are proposed by this document and become binding when John approves it.

Platform documents are drafted (Oct 9, 2026, awaiting John's approval) and are linked here: ../Platform/design-doc.md, ../Platform/adrs/, ../Platform/contracts/, ../Platform/openapi-serving-template.yaml. Project 1 and Project 2 documents are linked the same way.

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1 Status, owner, reviewers, last updated

Field	Value
Status	Draft, awaiting John's review and approval before coding of this project begins
Owner	John (GitHub: jwasikye)
Reviewers	John (approves each document) [DP: How the documents get drafted]
Last updated	Oct 9, 2026
Build position	Last of the five projects: Project 1, Project 3, Project 5, Project 2, then Project 4 [DP Purpose; BP How the plan works]

---

2 Summary

Project 4 tells a visitor tonight's chance of seeing a shooting star at a location and the top factor behind it. A nightly pipeline combines four inputs: NOAA GFS cloud cover (GFS is NOAA's Global Forecast System), the VIIRS nighttime lights composite (a satellite measure of light pollution), the Moon's altitude and illumination from the Skyfield library, and the International Meteor Organization (IMO) shower calendar. The pipeline turns them into a score for each cell (one hexagon of the H3 grid) and each night, using a documented formula that ADR-0010 proposes and John has not

yet approved. It names the top factor and publishes the result as Parquet files. A lookup page, built on Project 1's map and the shared serving template, answers from those files through AWS Lambda and DuckDB while the session box is off. If a nightly run is late, the page shows the old data with a visible late notice and the time of the last good run.

Phase A covers the lower 48 states; Alaska and Hawaii follow once the VIIRS latitude check allows. Phase B extends to every land cell the VIIRS data covers, adds pre-written explanation phrases, adds a backtest against known shower peaks that also tunes the weights, and adds a "did you see one?" vote stored in DynamoDB. The project uses no AI model on purpose: the score is a formula [ADR-0001].

---

3 Context, scope and constraints

3.1 Background facts (confirmed)

Fact	Source
GFS on AWS: bucket noaa-gfs-bdp-pds, region us-east-1, four runs a day; total cloud cover for the entire atmosphere in percent, hourly to 120 hours, about 28 km cells; NOAA asks for attribution	DP Answered by research: Cloud cover data
Skyfield uses JPL ephemerides and agrees with the U.S. Naval Observatory to within 0.00001 arcseconds	DP Answered by research: Moon calculation accuracy
EOG's VNL page says many VIIRS nighttime lights products are CC BY 4.0; the annual composite's licence is unverified. A free Earth Observation Group account is needed to download it, and EOG's register page says programmatic access is paid-only from June 1, 2026	EOG pages read Oct 9, 2026 (ADR-0014, OQ-23); BP 4A.1
The IMO site has a copyright notice and no reuse terms; store only factual fields and credit it	BP Risks table: Meteor shower calendar
The nightly run starts at 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern) on a spot instance (a cheaper AWS computer that AWS can take back at short notice). It takes about 45 minutes, with a hard maximum of 2 hours. When no spot instance is available, the night is skipped and the next run catches up	BD Defaults used; BP How the box is run
Everything that runs around the clock (Lambda, Fargate) writes only to always-on storage (S3, DynamoDB, CloudWatch, X-Ray)	BP Engineering standard 16

3.2 System boundary

Inside: the Project 4 pipeline (Airflow DAGs and Spark jobs on the session box), its Iceberg tables and published Parquet, the forecast endpoint, the lookup page, and in Phase B the vote endpoint and DynamoDB table.

Outside (linked, not repeated): the platform (CI, Terraform, session box, serving template, status page, incident record, Parameter Store, dev/live separation), Project 1's map component, Project 2's fireball data and usage-event definitions.

3.3 Constraints

- Budget under $50 a month, browser only; this project adds no model API cost [BD Budget; ADR-0001].
- Hard limits: every Spark job has a maximum number of rows read and written and a maximum runtime. A job over either limit stops before publishing and opens an incident. John sets the values from a measured size estimate [BP Hard limits table; OQ-05].
- Non-commercial portfolio use [BP Assumptions].
- All stored timestamps are in UTC. Only the page converts them to local time [BD Timestamps].
- Privacy rules: no raw IP addresses in usage events [BP 2B.4]. Votes follow Project 1's rules: anonymous phone ID, limits per visitor and per day, no internet addresses (decided by John, Oct 9, 2026; ADR-0009). Numbers are open [OQ-15].
- License and credit: see section 12.4.
- Stack fixed by the platform: Spark, Airflow, Iceberg with Glue, DuckDB serving, dbt, Great Expectations, React, FastAPI, API Gateway, DynamoDB, Terraform, Docker, k3s, GitHub Actions [BP Tools table; Technology by project].

---

4 Goals and non-goals

Goals

- G1. Give a visitor tonight's score and top factor for a place, with every source credited, with the session box off [BP 4A.3]. When data is late, show the old data with a visible late notice and the last good run time; never hide it (ADR-0011).
- G2. Make the score explainable with a documented formula, documented weights and a named top factor [BP 4A.2].
- G3. Keep data fresh every night and never publish bad data [BD Data freshness; BP Engineering standard 4].
- G4. Extend to every land cell the VIIRS data covers and state the limits of that data on the page [BP 4B.1].
- G5. Explain results in pre-written phrases whose values match the table [BP 4B.2].
- G6. Show where the score agrees and disagrees with known peaks, and collect "did you see one?" votes [BP 4B.3].

Non-goals

- No machine learning model, no language model call, no agent, no eval harness, no model card [ADR-0001; DP].
- No Snowflake marts, MetricFlow metrics, Superset dashboards or Grafana dashboards for this project. The Technology table marks none of them; OQ-12 asks John to confirm.
- No Kafka or streaming (Technology table has none).
- No per-request computation of the score: the page reads the published file [ADR-0007].
- No Phase C features (hourly updates, saved-location email) in this document set. Decided by John, Oct 9, 2026: Phase C stays unspecified and needs its own design (hourly updates conflict with the model where the session box is off most of the day); its done-when checks are missing and are written when Phase C is designed. They are listed in requirements.md as F-C1 and F-C2 only.
- No claim that a score predicts what a visitor will see, beyond what the backtest shows.

---

5 Requirements

Full text: requirements.md. Summary:

- Features: F-A1 inputs, F-A2 score, F-A3 lookup page (Phase A); F-B1 global coverage, F-B2 explanations, F-B3 backtest and feedback (Phase B); F-C1 hourly updates, F-C2 saved-location email (Phase C, no criteria recorded).
- Functional: FR-01 to FR-23 (Phase A), FR-30 to FR-42 (Phase B).
- Quality: QR-01 availability with the session box off; QR-02 cold start up to 10 seconds, stated on the page; QR-03 nightly freshness; QR-04 Moon correctness; QR-05 explainability; QR-06 cost limits; QR-07 privacy; QR-08 credit and licensing; QR-09 repeatability; QR-10 data quality gate; QR-11 spot reliability; QR-12 security; QR-13 observability; QR-14 change detection.
- Acceptance: the build plan's "Done when" for 4A.1, 4A.2, 4A.3, 4B.1, 4B.2, 4B.3, plus the first-run checks.

---

6 Architecture

6.1 System context (C4 level 1)

```
flowchart LR
  visitor(["`Visitor
looks up a place`"]):::person
  owner(["`John, owner
approves, operates`"]):::person
  subgraph B["Shooting star forecast"]
    sss["`Shooting star forecast
nightly sky score per cell
and public lookup page`"]:::sys
  end
  subgraph IN["External inputs"]
    gfs["`NOAA GFS on AWS open data
cloud cover`"]:::ext
    eog["`Earth Observation Group
```

```
VIIRS annual lights`"]:::ext
    imo["`International Meteor Organization
shower calendar`"]:::ext
    jpl["`JPL ephemeris and Horizons
Moon positions`"]:::ext
  end
  subgraph REUSE["Other projects"]
    p1["`Project 1
map and serving template`"]:::ext
    p2["`Project 2
fireball baseline (Phase B)`"]:::ext
  end
  visitor -->|"looks up, reports sighting (HTTPS)"| sss
  owner -->|"approves, operates (GitHub, browser)"| sss
  sss -->|"reads cloud cover (S3)"| gfs
  sss -->|"annual file downloaded by hand (ADR-0018)"| eog
  sss -->|"reads calendar (HTTPS)"| imo
  sss -->|"ephemeris; Moon check (HTTPS)"| jpl
  sss -->|"reuses map and template"| p1
  sss -->|"reads baseline"| p2
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
  classDef ext fill:#f1f5f9,stroke:#64748b,color:#0f172a,stroke-dasharray:4 3
  classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
  classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
  classDef sys fill:#ede9fe,stroke:#6d28d9,color:#0f172a
```

6.2 Containers (C4 level 2)

```
flowchart TB
  visitor(["Visitor"]):::person
  ext["`GFS, VIIRS, IMO, JPL
input sources`"]:::ext
  subgraph SSS["Shooting star forecast"]
    direction TB
    site["`Lookup page
React, TypeScript
S3 and CloudFront`"]:::cont
    api["`Forecast API
FastAPI on Lambda, DuckDB
API Gateway HTTP API`"]:::cont
    pub[("`Published Parquet
S3
sky_score_by_cell_night,
source_credits`")]:::store
    votes[("`Votes
DynamoDB, PITR on
feedback_votes (Phase B)`")]:::store
    box["`Nightly pipeline
Airflow and Spark on k3s
session box on a spot instance, 10:00 UTC`"]:::cont
    lake[("`Lake tables
Iceberg on S3, Glue catalog`")]:::store
    params[("`Parameter Store
Systems Manager, every key`")]:::store
    ci["`CI
GitHub Actions`"]:::cont
  end
  visitor -->|"uses (HTTPS)"| site
  site -->|"looks up, votes (HTTPS JSON)"| api
  api -->|"reads with DuckDB (S3)"| pub
  api -->|"writes votes (Phase B)"| votes
  box -->|"pulls inputs"| ext
  box -->|"writes and reads"| lake
  box -->|"publishes after gate passes"| pub
  box -->|"reads keys"| params
  ci -->|"deploys pipeline code"| box
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
  classDef ext fill:#f1f5f9,stroke:#64748b,color:#0f172a,stroke-dasharray:4 3
  classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
  classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
  classDef sys fill:#ede9fe,stroke:#6d28d9,color:#0f172a
```

6.3 Components of the nightly pipeline (C4 level 3)

```
flowchart LR
  subgraph NP["Nightly pipeline: ingest and conditions"]
    direction LR
    dag["`Airflow DAGs
shooting_star_nightly,
shooting_star_viirs_load`"]:::cont
    gfs["`GFS ingest
Python, Spark
cloud cover to H3 res 4`"]:::cont
    viirs["`VIIRS ingest
Python, Spark
lights to H3 res 8`"]:::cont
    imo["`Shower calendar loader
Python
factual fields only`"]:::cont
    cond["`Conditions builder
PySpark`"]:::cont
    moon["`Moon calculator
Skyfield
altitude, illumination`"]:::cont
    lake[("`Iceberg tables
conditions`")]:::store
  end
  dag -->|"runs"| gfs
  dag -->|"runs"| imo
  dag -->|"runs"| cond
  dag -->|"runs on annual load"| viirs
  cond -->|"calls"| moon
  cond -->|"writes"| lake
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
  classDef ext fill:#f1f5f9,stroke:#64748b,color:#0f172a,stroke-dasharray:4 3
  classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
  classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
  classDef sys fill:#ede9fe,stroke:#6d28d9,color:#0f172a
```

```
flowchart LR
  subgraph NP2["Nightly pipeline: score, tests, gate and publish"]
    direction LR
    cond["Conditions builder"]:::cont
    score["`Score engine
Python, PySpark
score, top factor`"]:::cont
    dbt["`dbt project
dbt on DuckDB
stages, tests, mart`"]:::cont
    gate["`Publish gate
Great Expectations
freshness, ranges`"]:::cont
    pubr["`Publisher
Python
Parquet to published prefix`"]:::cont
    dag["Airflow DAGs"]:::cont
    metrics["`Run metrics emitter
freshness, row counts`"]:::cont
  end
  cond -->|"calls"| score
  score -->|"feeds"| dbt
  dbt -->|"feeds"| gate
  gate -->|"allows"| pubr
  dag -->|"runs last tasks"| metrics
  classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
```

6.4 Structure of the repository

The platform repository has the folders platform, pipelines, dbt, checks, app and docs [BP 0.1]. Project 4 puts its code in a subfolder named shooting_star (a proposed name) inside each of those folders.

---

7 Component breakdown

Function, class, file and table names below are proposed. The Documentation Plan names four especially tricky parts that get a full function-by-function listing: terrain horizon (1A.2), satellite windows (1A.3), vessel identity matching (3B.3 and 3B.4) and transcript time alignment (5A.4). None of them belongs to Project 4, so every component here lists only its main functions and classes. Types are written Python-style and are descriptive; the code fixes the final signatures.

7.1 Orchestration (Airflow DAGs): pipelines/shooting_star/dags/

- Job: Airflow (the scheduler) runs the tasks in order, retries failed tasks, runs backfills (reruns for past dates) and writes the run metrics. A DAG is one scheduled pipeline definition.
- Inputs: the nightly schedule (10:00 UTC), the run's data interval.
- Outputs: task states, metrics rows, the published tables.
- Main items:
- DAG `shooting_star_nightly`: tasks in order: `ingest_gfs`, `load_shower_calendar`, `compute_conditions` (calls the Moon calculator and the conditions builder), `compute_score`, `dbt_build`, `publish_gate`, `publish`, `emit_metrics`, and, last, the Iceberg maintenance task that the platform provides [BP Engineering standard 7]. Iceberg is the table format used on S3.
- DAG `shooting_star_viirs_load`: reads the VIIRS annual file that John downloads by hand once a year (ADR-0018), and writes the manifest and the cells (ADR-0012). The yearly re-download is decided (ADR-0012, decided by John, Oct 9, 2026); only its exact schedule or trigger is open [OQ-07].
- Functions: `data_interval_to_night(logical_date) -> night_date` (local date of the sunset, ADR-0013), `on_failure_callback(context)` (platform incident hook).
- Notes: Airflow reads its connection keys from Parameter Store through its built-in secrets backend [BP 0.3]. The retry counts and timeouts are open [OQ-20].

7.2 GFS ingest: pipelines/shooting_star/ingest/gfs_ingest.py

- Job: download that morning's GFS total cloud cover for the entire atmosphere (hourly, out to 120 hours), record each file in the raw manifest (a table of URL, fetch time and hash), convert the grid values to H3 resolution 4 cells, and write them to the Iceberg table.
- Inputs: model run time and forecast hours (OQ-04); S3 bucket noaa-gfs-bdp-pds (us-east-1).
- Outputs: `gfs_cloud_cover_raw_manifest` rows; `gfs_cloud_cover_h3` partition for the run (contracts/gfs_cloud_cover.yaml).
- Main functions:
- `select_gfs_run(now_utc, lag) -> model_run_time_utc` chooses the run to read (rule open, OQ-04).
- `list_gfs_files(model_run_time_utc, hours) -> list[S3Object]`.
- `download_cloud_file(s3_object) -> LocalFile` downloads the file, computes its hash and records its size.
- `read_cloud_grid(local_file) -> Grid` reads the file in the format that the first-run check confirms (OQ-04).
- `grid_to_h3_r4(grid) -> DataFrame` maps 0.25 degree grid points to H3 resolution 4 cells. How several grid points in one cell combine is open (OQ-06).
- `write_gfs_partition(df, model_run_time_utc)` overwrites the partition (the slice of the table) for that run.
- Class: `GfsRun` (dataclass: run time, forecast hours, file list).

7.3 VIIRS ingest: pipelines/shooting_star/ingest/viirs_ingest.py

- Job: read the annual composite that John downloads by hand with a free EOG login (ADR-0018), record it in the manifest, convert the raster (a grid of pixel values) to H3 resolution 8 cells, and write the cells to Iceberg.
- Inputs: product year and file (OQ-07). The file is downloaded by hand with a free EOG login, and no credential is stored (ADR-0018). EOG limits programmatic access to paid subscribers from June 1, 2026 (OQ-23).
- Outputs: `viirs_raw_manifest` rows; `viirs_light_h3` for the composite year (contracts/viirs_nightlights.yaml).
- Main functions:

- `verify_viirs_composite(local_file, year) -> ManifestRow` checks the hand-downloaded file and computes its size and hash (ADR-0018).
- `read_raster(local_file) -> Raster`.
- `raster_to_h3_r8(raster) -> DataFrame` combines the pixels inside each cell into one value. The combining rule is open (OQ-07).
- `write_viirs_partition(df, year)`.
- `covered_latitude_range(raster) -> (min_lat, max_lat)` returns the latitude limits of the file. This answers OQ-01 and feeds the Phase B page notice.
- Class: `ViirsComposite` (dataclass: year, source URL, hash, latitude range).

7.4 Shower calendar loader: pipelines/shooting_star/ingest/imo_calendar.py

- Job: retrieve the IMO calendar and keep only the fields the score needs: shower name, active dates, peak date and rate [ADR-0005].
- Inputs: the IMO calendar; its format and URL are open (OQ-14).
- Outputs: `imo_shower_calendar` and its manifest (contracts/imo_shower_calendar.yaml).
- Main functions: `fetch_calendar() -> RawCalendar`; `parse_calendar(raw) -> list[ShowerRow]` (keeps only the fields the score needs, each with its source URL; drops all others); `write_calendar(rows)`.
- Class: `ShowerRow` (dataclass: shower_name, active_start, active_end, peak_date, rate, source_url).

7.5 Moon calculator: pipelines/shooting_star/moon/moon_calculator.py

- Job: compute Moon altitude and illumination with Skyfield and a JPL ephemeris [ADR-0003].
- Inputs: ephemeris file (contracts/jpl_ephemeris.yaml), cell centre latitude and longitude, and the hourly sample times between sunset and sunrise of the night (ADR-0013).
- Outputs: for each cell and night, the mean Moon illumination, the highest Moon altitude and the Moon glare value (see ADR-0010), as the `MoonNight` class.
- Main items:
- Class `MoonCalculator(ephemeris_path)` with `load()`, `altitude_deg(lat, lon, t_utc) -> float`, `illumination_pct(t_utc) -> float`, `sun_events(lat, lon, local_date, tz) -> (sunset_utc, sunrise_utc)` (proposed to use Skyfield; the time zone source is OQ-03), `night_summary(lat, lon, night) -> MoonNight` (hourly samples between sunset and sunrise, see ADR-0010).
- Function `moon_udf` wraps the class so the Spark job can call it (a UDF is a user-defined function inside Spark).
- Function `horizons_check(positions) -> list[Delta]` returns the differences from JPL Horizons for the Phase B test. The reference values and the tolerance are open (OQ-09).
- Class: `MoonNight` (dataclass: illumination_pct_mean, altitude_max_deg, glare).

7.6 Conditions builder: pipelines/shooting_star/jobs/build_conditions.py (PySpark)

- Job: assemble the per-cell, per-night conditions by joining four things: cloud cover (read from the resolution 4 parent cell), light (resolution 8), Moon and shower.
- Inputs: `gfs_cloud_cover_h3`, `viirs_light_h3`, `imo_shower_calendar`, Moon values, the night.
- Outputs: the `conditions_by_cell_night` table in Iceberg (internal; never published directly). Proposed split of work: Spark computes the conditions and the score (this job and 7.7); dbt stages and tests them and builds the published table (7.8).
- Main functions:
- `build_conditions(spark, night_date, model_run_time_utc) -> DataFrame`.
- `attach_cloud(cells_df, cloud_df)` using `cell_to_parent(cell_h3_r8, 4)` [ADR-0004].
- `attach_moon(cells_df, moon_udf)`.
- `attach_shower(cells_df, calendar_df, night_date)` computes `shower_name` and `shower_activity` as in ADR-0010.
- `enforce_job_limits(df, max_rows, max_runtime)` stops the job before publishing when a limit is exceeded, and opens an incident [BP Hard limits table]. The limit values are open (OQ-05).

- The job writes one partition per night and overwrites it on a rerun (engineering standard 3 in the build plan).

7.7 Score engine: pipelines/shooting_star/score/score_engine.py

- Job: compute the score and the top factor from the conditions, using documented weights [BP 4A.2; ADR-0001].
- Inputs: one conditions row (cloud_cover_pct, moon_illumination_pct, moon_altitude_max_deg, moon_glare, light_value, light_rank, shower_name, shower_activity) and a `ScoreConfig`.
- Outputs: `score`, `top_factor`, `score_version`, written by a Spark job to the `score_by_cell_night` Iceberg table. In the proposed split, Spark computes the score, and dbt only stages and tests it and builds the published table from it.
- Main items:
- Class `ScoreConfig` (holds the weights moon 0.40, light 0.40 and shower 0.20 as STARTING VALUES, plus `score_version`; stored in the repository and versioned).
- `factor_values(row) -> FactorValues` (clear_sky, moon_dark, dark_sky, shower_activity).
- `compute_score(values, config) -> float`.
- `points_lost(values, config) -> dict[Factor, float]`.
- `top_factor(points_lost) -> Factor` (tie order cloud_cover, moon, light_pollution, no_active_shower).
- `score_dataframe(df, config) -> DataFrame` applies the same functions inside Spark, either as column expressions or as a UDF, so that the unit tests and the Spark job give the same results. Which of the two is used is an implementation detail decided in the code.
- Enum `Factor`: cloud_cover, moon, light_pollution, no_active_shower.
- John approved ADR-0010 on Oct 9, 2026, so this component may be built.

7.7.1 Score specification (APPROVED)

The formula, the ingredients, the starting weights, the night sampling and the top-factor rule are written once, in ADR-0010. That ADR was approved by John on Oct 9, 2026, as written (OQ-02). This design doc does not repeat the formula, so the two cannot disagree. The weights are STARTING VALUES and are not validated.

What is not decided in ADR-0010: how cloud values are assigned to cells and what the score grain is (OQ-06), the time zone source and the sunrise and sunset method (OQ-03), the VIIRS unit (OQ-07), and whether the IMO calendar provides active dates (OQ-14). The formula does not use the radiant position of a shower (the point in the sky it appears to come from), so the position is not stored.

7.8 dbt project: dbt/shooting_star/

- Job: build the dbt models on DuckDB in three layers (staging, intermediate and marts) with tests [BP 4A.2, Engineering standard 6]. A mart is a final table meant for use. In the proposed split, Spark writes the input, conditions and score tables to Iceberg, and dbt stages them, tests them and builds the published mart. dbt does not recompute the score. Two points stay open: how dbt on DuckDB reads Iceberg, and whether marts stay on DuckDB only (OQ-12).
- Inputs: Iceberg input, conditions and score tables (access method from the platform serving/dbt setup).
- Outputs: marts `sky_score_by_cell_night` and `source_credits`.
- Models (proposed): `stg_gfs_cloud_cover`, `stg_viirs_light`, `stg_imo_showers`, `stg_moon`, `stg_score_by_cell_night`, `int_conditions_by_cell_night`, `sky_score_by_cell_night`, `source_credits`.
- Tests: unique and not_null tests on the keys (`cell_h3` plus `night_date`, and `source_id`), an accepted-values test for `top_factor`, and range tests for the percent columns.
- Phase B adds the dbt docs catalog page (a generated website that describes the models) and OpenLineage events (records of which data each job read and wrote) [BP Technology table].

7.9 Publish gate: checks/shooting_star/

- Job: run Great Expectations checks (a data quality tool) on forecast freshness and value ranges. The gate blocks publishing when a check fails [BP 4A.2, Engineering standard 4].
- Inputs: the staged `sky_score_by_cell_night`, the run time, the freshness threshold (OQ-20).
- Outputs: a pass or fail result with a results record. A failure opens an incident.

- Main items: suites `forecast_freshness` and `sky_score_ranges`; function `run_publish_gate(table, run_context) -> GateResult`; sample files for CI tests (a passing set and a failing set).

7.10 Publisher: pipelines/shooting_star/publish/publisher.py

- Job: once the gate passes, write the final (gold) tables as Parquet files to the published location in S3, replacing only that night's partition.
- Inputs: staged tables, `GateResult`.
- Outputs: Parquet files that the Lambda function reads.
- Main functions: `publish_tables(staged, gate_result)` (refuses to publish unless `gate_result` is a pass); `write_manifest(published_files)`.

7.11 Run metrics emitter: pipelines/shooting_star/ops/metrics.py

- Job: write the freshness and row counts of each task to the platform metrics table, so that the status page shows the last good run [BP 0.5].
- Main function: `emit_run_metrics(run_id, table_name, rows, freshest_event_time)`.

7.12 Forecast API: app/shooting_star/api/ (FastAPI on Lambda)

- Job: answer lookups by querying the published Parquet files with DuckDB; in Phase B, also accept votes.
- Inputs: HTTP requests per openapi.yaml.
- Outputs: JSON responses, and in Phase B vote items in DynamoDB.
- Main items:
- `app = FastAPI()` with routes `get_forecast(lat, lon)` and (Phase B) `post_vote(body)`.
- `point_to_cell(lat, lon, resolution) -> cell_h3` (uses the h3 library; resolution OQ-06).
- `tonight_night_date(lat, lon, now_utc) -> night_date` (local night at the place, ADR-0013; time zone source OQ-03).
- `query_forecast(cell_h3, night_date) -> ForecastRow | None` runs DuckDB over the Parquet and returns the row for `night_date`, or the latest earlier row when tonight has none (ADR-0011).
- `load_credits() -> list[Credit]`.
- `build_response(row, credits, tonight) -> ForecastResponse` sets `late` when `row.night_date < tonight` and `last_good_run_time_utc` from `row.loaded_at_utc`.
- `render_explanation(row) -> str` (Phase B; see 7.14).
- Pydantic models (Python classes that validate request and response shapes): `ForecastResponse`, `Conditions`, `PointsLost`, `Credit`, `NoData`, `VoteRequest`, `Error`.
- The platform's serving template decides how FastAPI runs on Lambda [../Platform/design-doc.md].

7.13 Lookup page: app/shooting_star/web/ (React and TypeScript)

- Job: let the visitor enter a place, and show the answer, the source credits and any notices.
- Built from the serving template's map component, lookup form, answer card and status page, and Project 1's map [ADR-0008].
- Main components (proposed): `LookupForm`, `ScoreCard`, `FactorBadge`, `ConditionsList`, `SourceCredits`, `ColdStartNotice` (the wording is in section 11 and BP 0.4), `LateNotice` (always visible when data is late; shows the time of the last good run in local time; never hides the data), `OutsideCoverageNotice`, `TimeText` (UTC to local), and in Phase B `LatitudeLimitNotice`, `ExplanationText`, `SawOneButton` (the phone computes the H3 cell and keeps the random ID).
- Main functions: `fetchForecast(lat, lon)`, `postVote(vote)`, `formatLocalTime(utc, tz)`.
- Place entry and geocoding: OQ-10.

7.14 Explanation phrases (Phase B): app/shooting_star/phrases/

- Job: hold the pre-written phrases and fill them with the computed values [ADR-0002].
- Inputs: a score row.

- Outputs: one sentence for the page.
- Main items: `PHRASES` (a map from each top factor and combination to a sentence template); `render_phrase(row) -> str`; test `test_phrase_values_match_table` (checks that every value in a rendered phrase equals the value in the score table).

7.15 Backtest and Moon check (Phase B): pipelines/shooting_star/backtest/

- Job: compare the scores with known shower peaks and with Project 2's fireball baseline, and check 20 Moon positions against JPL Horizons [BP 4B.3].
- Inputs: `sky_score_by_cell_night`, `imo_shower_calendar`, Project 2's fireball table (../Asteroid Watch/contracts/), reference Horizons positions (OQ-09).
- Outputs: a short note that states where the score agrees and disagrees with known peaks, and the Moon test results.
- Main functions: `compare_with_peaks(scores, calendar) -> PeakComparison`; `compare_with_fireballs(scores, fireballs) -> FireballComparison`; `moon_vs_horizons(positions, tolerance) -> list[Delta]`; `write_note(comparisons) -> markdown`. The comparison methods are open (OQ-16).

7.16 Votes (Phase B): app/shooting_star/api/votes.py and Platform/terraform/

- Job: store each vote in the DynamoDB table and count the votes [ADR-0009].
- Main functions: `store_vote(vote) -> None` (checks and updates the separate limit counters, stores no visitor ID with the vote); `count_votes(night_date=None) -> int`. The limit numbers and other details are open (OQ-15).
- Terraform: one dev table and one live table, each with provisioned capacity within the free tier. Point-in-time recovery (restoring the table to an earlier moment) follows OQ-15; the platform default is on.

7.17 Infrastructure: Platform/terraform/ (project module)

- Resources this project uses, built from platform modules: S3 locations for raw, lake and published data; one Glue database (the Iceberg catalog) per environment; a Lambda function; an API Gateway HTTP API; a CloudFront site; one least-privilege IAM role per pipeline; Parameter Store entries (none for VIIRS, because the download is by hand; ADR-0018); and a DynamoDB table in Phase B. The platform design doc gives the names, sizes and limits [../Platform/design-doc.md].

---

8 Runtime view

8.1 Nightly run (10:00 UTC)

```
sequenceDiagram
  participant S as Scheduler
  participant B as Session box (spot)
  participant A as Airflow
  participant G as GFS (S3)
  participant L as Lake and published S3
  S->>B: Start at 10:00 UTC
  B->>A: Restore Airflow database from S3
  A->>G: ingest_gfs
  A->>L: Write raw manifest and Iceberg partition
  A->>A: load_shower_calendar, compute_conditions
  A->>A: compute_score, dbt_build
  A->>A: publish_gate
  alt gate passes
    A->>L: publish Parquet
  else gate fails
    A->>A: stop, incident, email
  end
  A->>L: emit_metrics, Iceberg maintenance
  A->>B: Shut down when the pipelines finish (max 2 hours)
```

Steps and failure handling:

1. The session box starts on a spot instance. If none is available, the night is skipped and the next run catches up [BP How the box is run]. No new score exists for the skipped night; the page keeps showing the latest earlier row with a late notice and the last good run time (ADR-0011).
2. The platform restores Airflow's database from S3, and Airflow reads its keys from Parameter Store [BP 0.3].
3. `ingest_gfs` selects the GFS run (OQ-04), downloads the files, records the URL, fetch time and hash, and writes the partition. If a file is missing or unreadable, the task fails. The retry counts and timeouts are open (OQ-20). A failure ends in an incident and an email [BP 0.5].
4. Before loading, each task compares the schema of its source with the contract. A mismatch fails the task before any write (engineering standard 5).
5. `compute_conditions` and `compute_score` run under the maximum-rows and maximum-runtime limits. A job that goes over either limit stops before publishing and opens an incident [BP Hard limits table].
6. `dbt_build` runs the models and tests. A failing test fails the task.
7. `publish_gate` checks forecast freshness and value ranges. When a check fails, three things happen: nothing is published, earlier published nights stay as they are, and an incident is opened. A lookup is then served the latest earlier row with the late notice (ADR-0011, decided by John, Oct 9, 2026). The partition for a night is overwritten only after success, so a rerun is safe (engineering standard 3).
8. `emit_metrics` writes the freshness and row counts, so that the status page shows the last good run.
9. The Iceberg maintenance task (clean-up of small files and old metadata) is the last task of the nightly run, so it never competes with the pipelines [BP Engineering standard 7].
10. The session box shuts down when the pipelines finish. Reaching the 2-hour maximum is recorded as an incident [BD Nightly shutdown].
11. If AWS takes back the spot instance that hosts the session box, the platform's interruption handler reacts to the two-minute warning, and every task is safe to rerun [BP How the box is run].

8.2 Lookup request (session box off)

```
sequenceDiagram
  participant V as Visitor browser
  participant C as CloudFront and S3
  participant G as API Gateway
  participant F as Lambda (FastAPI, DuckDB)
  participant P as Published Parquet
  V->>C: Load the page
  V->>V: Enter a place (OQ-10)
  V->>G: GET /forecast?lat&lon
  G->>F: Invoke
  F->>F: point_to_cell
  F->>P: DuckDB query for the cell, tonight or the latest earlier night
  P-->>F: Row or nothing
  F-->>V: 200 with score, top factor, credits (late set if tonight has no row), or 404 NoData
```

Failure handling:

- Invalid coordinates return 400.
- A point with no published data at all (outside coverage) returns 404 with a reason.
- A missing tonight does not return 404. The endpoint returns the latest earlier row with `late` set to true, and the page shows the late notice (ADR-0011).
- A cold start (the first request after inactivity) may take up to 10 seconds, and the page says so [BP 0.4].
- A Lambda error returns 500, and the page shows an error message.
- API Gateway throttling returns 429. The throttle values are open (OQ-19).

8.3 Vote (Phase B)

The page sends POST /votes with the cell that the phone computed and the random phone ID. The Lambda function checks the separate limit counters, writes the vote without the ID to DynamoDB, and returns 202. A request over the limit returns 429. A failed write returns 500 and nothing is stored. The limit numbers and other details are open (OQ-15).

8.4 VIIRS annual load

`shooting_star_viirs_load` runs once a year (ADR-0012): it reads the annual file that John downloads by hand with a free EOG login (ADR-0018), writes the manifest (URL, size, hash) and the resolution 8 cells, and records the file's covered latitude range. Only the manifest is kept, not the raw file. The yearly re-download is decided; its exact schedule or trigger is open (OQ-07). A failed download leaves the previous year's table in use.

8.5 Change flow

On every merge, GitHub Actions runs the changed pipelines with Spark on sample data, without starting the AWS session box. Only if they pass does the change deploy to dev and then to live. Every site change also gets a preview link [BP How the plan works].

8.6 Data flow

```
flowchart TB
  subgraph SRC["Sources"]
    gfs["`NOAA GFS
cloud cover`"]:::ext
    eog["`Earth Observation Group
VIIRS lights`"]:::ext
    imo["`IMO
shower calendar`"]:::ext
    jpl["JPL ephemeris"]:::ext
  end
  subgraph RAW["Raw manifests (S3)"]
    mg[("gfs_cloud_cover_raw_manifest")]:::store
    mv[("viirs_raw_manifest")]:::store
    mi[("imo_shower_calendar_raw_manifest")]:::store
    mj[("jpl_ephemeris_manifest")]:::store
  end
  subgraph LAKE["Lake tables (Iceberg)"]
    lg[("gfs_cloud_cover_h3")]:::store
    lv[("viirs_light_h3")]:::store
    li[("imo_shower_calendar")]:::store
    mo["`moon_nightly
(Skyfield)`"]:::cont
    lc[("conditions_by_cell_night")]:::store
    ls[("score_by_cell_night")]:::store
  end
  subgraph PUB["Published (Parquet on S3), after dbt build and publish gate"]
    ps[("sky_score_by_cell_night")]:::store
    pc[("source_credits")]:::store
  end
  api["`Forecast API
Lambda, DuckDB`"]:::cont
  site["Lookup page"]:::cont
  gfs --> mg --> lg
  eog --> mv --> lv
  imo --> mi --> li
  jpl --> mj --> mo
  lg --> lc
  lv --> lc
  li --> lc
  mo --> lc
  lc --> ls
  ls --> ps
  ps --> api
  pc --> api
  api --> site
  classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
  classDef ext fill:#f1f5f9,stroke:#64748b,color:#0f172a,stroke-dasharray:4 3
  classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
  classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
  classDef sys fill:#ede9fe,stroke:#6d28d9,color:#0f172a
```

---

9 Data model

Contracts are in contracts/ in ODCS format. Platform shared tables (metrics, incidents) are in ../Platform/contracts/.

Table	Layer	Phase	Contract	Notes
gfs_cloud_cover_raw_manifest	raw manifest	A	gfs_cloud_cover.yaml	URL, fetch time, hash
gfs_cloud_cover_h3	Iceberg, input table	A	gfs_cloud_cover.yaml	H3 resolution 4, per run and valid time
viirs_raw_manifest	raw manifest	A	viirs_nightlights.yaml	manifest only (URL, size, hash), no raw file; re-download once a year (ADR-0012)
viirs_light_h3	Iceberg, input table	A	viirs_nightlights.yaml	H3 resolution 8, per composite year
imo_shower_calendar_raw_manifest, imo_shower_calendar	raw manifest, reference	A	imo_shower_calendar.yaml	only the fields the score needs, each with its source (ADR-0005)
jpl_ephemeris_manifest	raw manifest	A	jpl_ephemeris.yaml	file used by Skyfield
moon_nightly (internal, computed)	Iceberg, intermediate	A	not a source and not published, so it has no contract; its columns are in the sky_score contract	Moon illumination, altitude and glare per cell and night
conditions_by_cell_night (internal)	Iceberg, written by Spark	A	columns are in the sky_score contract	cloud, light, Moon, shower
score_by_cell_night (internal)	Iceberg, written by Spark	A	columns are in the sky_score contract	score, top_factor, score_version
sky_score_by_cell_night	published Parquet (dbt mart built from the Spark score table)	A	sky_score_by_cell_night.yaml	key (cell_h3, night_date)
source_credits	published Parquet	A	source_credits.yaml	four rows in Phase A
feedback_votes, feedback_vote_limits	DynamoDB	B	feedback_votes.yaml	votes carry no visitor ID; counters stored apart and expiring (ADR-0009); numbers OQ-15

Published tables: entity-relationship view

```
erDiagram
  sky_score_by_cell_night {
    string cell_h3 PK
    date night_date PK
    timestamp night_start_utc
    timestamp night_end_utc
    timestamp forecast_run_time_utc
    number cloud_cover_pct
    number moon_illumination_pct
    number moon_altitude_max_deg
    number moon_glare
    number light_value
    number light_rank
    string shower_name
    number shower_activity
    number score
    string top_factor
    number lost_cloud_cover
    number lost_moon
    number lost_light_pollution
    number lost_shower
    string score_version
    timestamp loaded_at_utc
  }
  source_credits {
    string source_id PK
    string display_name
```

```
    string credit_line
    string terms_note
    string source_url
  }
  feedback_votes {
    string vote_id PK
    date night_date
    string cell_h3
    boolean saw_one
    timestamp created_at_utc
    timestamp expires_at_utc
  }
  gfs_cloud_cover_h3 {
    string cell_h3_r4 PK
    timestamp valid_time_utc PK
    timestamp model_run_time_utc PK
    number total_cloud_cover_pct
  }
  viirs_light_h3 {
    string cell_h3_r8 PK
    integer composite_year PK
    number radiance_value
  }
  imo_shower_calendar {
    string shower_name PK
    date peak_date PK
    date active_start
    date active_end
    number rate
  }
  gfs_cloud_cover_h3 ||--o{ sky_score_by_cell_night : "cloud via cell_to_parent"
  viirs_light_h3 ||--o{ sky_score_by_cell_night : "light value"
  imo_shower_calendar |o--o{ sky_score_by_cell_night : "shower activity"
  sky_score_by_cell_night |o--o{ feedback_votes : "cell_h3, night_date (Phase B)"
```

The relationships are logical joins made by the nightly run, not database foreign keys. source_credits is returned alongside every answer and has no key link to the score table.

Raw-file Object Lock (30 days, live raw only) and S3 clean-up rules are defined by the platform: ../Platform/design-doc.md.

Rules applied: raw immutable with URL, fetch time and hash (engineering standard 1); event time and source key on every record (2); partition overwrite, never blind append (3); H3 resolutions per ADR-0004; UTC timestamps (15). History: the build plan lists no slowly changing history (SCD Type 2, which keeps old versions of changing rows) for Project 4, so none is designed.

---

10 API

Specification: openapi.yaml. Provisional paths and shapes; to be aligned with the serving template ../Platform/openapi-serving-template.yaml.

Method and path	Phase	Purpose	Responses
GET /forecast	A	Score and top factor for a point (tonight, or the latest earlier night with `late` set), with credits	200, 400, 404 (no data for the cell), 429, 500
POST /votes	B	Store a "did you see one?" answer	202, 400, 429, 500

API type: API Gateway offers an HTTP type, used for public pages, and a REST type, which supports API keys and usage plans [BP Tools table]. Project 4 uses the HTTP type because no key requirement is recorded for it. The rate limits, throttle values and caching values are not recorded (OQ-19).

---

11 Deployment view

What	Runs where	Notes
Nightly DAGs, Spark jobs, dbt, publish gate	Session box on k3s (t4g.2xlarge everyday; m7g.2xlarge for heavy batch), spot only, started 10:00 UTC, shut down when done, 2-hour maximum	[BP 0.3, How the box is run]; which server type this project's job needs is open (OQ-05)
Lookup endpoint	Lambda (FastAPI, DuckDB) behind API Gateway HTTP API	works with the session box off
Page	Static files on S3 and CloudFront	preview link per pull request
Published Parquet, lake tables	S3 (Iceberg with Glue catalog)	dev and live fully separate buckets, Glue databases and roles; dev cannot write live data [BP 0.2]
Votes (Phase B)	DynamoDB; point-in-time recovery follows OQ-15	dev and live separate
Keys	Parameter Store	No EOG credential: the VIIRS file is downloaded by hand (ADR-0018)
CI, deploys	GitHub Actions with OIDC	Spark tests on sample data; then dev; then live
Editing, small tests	GitHub Codespaces	[BP Where each job runs]
Infrastructure	Terraform; state in the platform's dedicated bucket	[BD Terraform state]

Storage rules: live raw files are locked against deletion or overwrite for 30 days (dev files are not), and S3 clean-up rules apply to every bucket; both are platform rules. They are described in ../Platform/design-doc.md (storage and retention) and are not repeated here.

Schedules: nightly 10:00 UTC; VIIRS load once a year (decided, ADR-0012; exact schedule or trigger OQ-07); weekly smoke test of sources (platform step 1B.9 includes every source a project uses).

Cold-start wording on the page (from BP 0.4): "The first request after a period of inactivity may take up to 10 seconds while the service initializes."

---

12 Cross-cutting concerns

12.1 Security

Least privilege means each pipeline gets only the access it needs. In practice: one IAM role per pipeline, read-only where possible; dev roles cannot write live data; every key in Parameter Store and none in code or GitHub; gitleaks (secret scanner) and Trivy (image scanner) in CI; source text treated as data and never as instructions, which matters for the IMO page [BP Engineering standard 13, 0.1, 0.2]. The threat model is written while building, in the security and privacy document.

12.2 Privacy

No raw IP addresses in usage events [BP 2B.4]. Timestamps in UTC. Votes follow Project 1 (decided by John, Oct 9, 2026): anonymous random phone ID kept on the phone, limits per visitor and per day, no internet addresses, cell computed on the phone, counters stored apart from votes [ADR-0009]; numbers are open [OQ-15]. Lookup coordinates sent to the API: privacy is open [OQ-10]. A personal-data inventory is written while building.

12.3 Observability

Freshness and row counts per run to the metrics table; status page; email on a missed freshness target or failed run; one incident record per real incident; one de-duplicated GitHub issue per failure [BP 0.5, 1B.9, Engineering standard 11]. OpenLineage and the dbt docs catalog in Phase B.

12.4 Licensing and credit

- GFS: NOAA asks for attribution [DP].
- VIIRS: many VIIRS nighttime lights products are CC BY 4.0 according to EOG. The annual composite's licence is unverified, and the build plan's public-domain claim is not confirmed (ADR-0014).

- IMO: copyright notice, no reuse terms; factual fields only; credit [BP Risks table; ADR-0005].
- JPL ephemeris: the SSD page has no licence statement; cite Park et al. 2021 (ADR-0014). The public-domain statement from a 2022 JPL email is UNVERIFIED.
- Skyfield: MIT licence (ADR-0014).
- Each input has a terms note and a credit line shown on the page (`source_credits`).
- Cloud cover cells are about 28 km wide, so the page should not suggest finer cloud detail than that (ADR-0004).
- Non-commercial portfolio; check licensing before any charging [BP Assumptions].

12.5 Cost limits

- Every Spark job: maximum rows and runtime; values from the measured size estimate [OQ-05].
- Nightly maximum 2 hours; 4-hour session limit and idle shutdown; $40 AWS budget as backup [BP Hard limits table].
- DynamoDB provisioned within the free tier; heavy traffic is throttled, not billed.
- Lambda and API Gateway HTTP API costs follow the platform's cost table in the build plan.
- No model API cost (ADR-0001).
- S3 clean-up rules and storage-size alarm from platform step 1B.9.

12.6 Reliability

Reliability rests on four measures: idempotent partition overwrite, write-audit-publish, the spot interruption handler, and Iceberg maintenance as the last task. A skipped or failed night leaves no new score for that night; the page shows the latest earlier row with a late notice and the last good run time (ADR-0011).

12.7 Time zones

All stored timestamps are UTC; the page converts to local time [BP Engineering standard 15]. A night is the local night at the place, sunset to sunrise (ADR-0013); the time zone source is OQ-03.

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13 Testing strategy

Level	What	Where
Unit	`compute_score`, `top_factor` and `points_lost` with fixed rows; `MoonCalculator` with fixed times; `parse_calendar` with a saved sample; `point_to_cell`; `select_gfs_run`	CI
Pipeline	The changed Spark jobs run on small sample files with Spark on GitHub's free CI machines (the AWS session box stays off); a rerun gives the same rows (idempotence)	CI
Data tests	dbt tests (unique, not_null on keys, accepted values, ranges); Great Expectations suites tested with a passing and a failing sample set so publishing stops on failure	CI
Contract	Schema diff against each ODCS contract before loading	CI and nightly run
Limits	A job given a limit below its input stops before publishing	CI
Moon	The one 20-position check against JPL Horizons within a stated tolerance (OQ-09); it is both the Phase B first-run check and the 4B.3 test	CI, Phase B
Phrases	Every rendered phrase value equals the score table value	CI, Phase B
Score behavior	A full-Moon night and a night near a shower peak give clearly different scores and top factors (written as in the plan; a numeric margin is set after John approves the formula, ADR-0010, decided by John, Oct 9, 2026)	CI with fixtures
Late data	With tonight's row absent, the API returns the latest earlier row with `late` true and `last_good_run_time_utc`, and the page shows the late notice	CI with fixtures
Score arithmetic	The four points-lost values add up to 100 minus the score; the top factor follows the tie order	CI
API	Contract tests of GET /forecast against openapi.yaml; 400 and 404 (no data) cases	CI
End to end	With the session box off, the public link answers a lookup	After each deploy to live (automation: OQ-19)

Level	What	Where
First-run checks	GFS files read correctly from AWS; VIIRS download works with a free EOG account (Phase A; see OQ-23); the 20-position Moon check is the 4B.3 test, listed in the Moon row above (Phase B)	Before building on each assumption
Weekly	A smoke test (a quick check that the source still works) pulls one small sample from every source used; how it samples VIIRS is open (OQ-07)	Scheduled

Evals (tests that score model answers), an LLM judge and red-team sets (prompts written to attack a model) do not apply, because the project has no model.

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14 Alternatives considered

Decision	Options weighed	Record
Score method	Formula (chosen) vs a trained model	ADR-0001
Explanation text	Pre-written phrases (chosen) vs generated text	ADR-0002
Moon calculation	Skyfield with a JPL ephemeris; Horizons used as a test only	ADR-0003
Grid resolutions	H3 resolution 4 for GFS, 8 for VIIRS, joined with cell_to_parent	ADR-0004
Score formula	Formula with starting weights (approved Oct 9, 2026) vs other weights, cloud as a sum term, an absolute light scale	ADR-0010
Shower calendar	Fields the score needs, each with its source (chosen) vs copying more	ADR-0005
Coverage	Lower 48 first, Alaska and Hawaii after the VIIRS check, then global (chosen) vs global from the start	ADR-0006
Refresh	Nightly run (chosen) vs weekly; hourly deferred to Phase C	ADR-0007
Page	Reuse Project 1 map and serving template	ADR-0008
Late data	Show old data with a late notice (chosen) vs hide it vs 404	ADR-0011
VIIRS storage	Manifest and yearly re-download (chosen) vs a full raw copy	ADR-0012
Meaning of tonight	Local night, sunset to sunrise	ADR-0013
Votes	DynamoDB (chosen) vs PostgreSQL on the session box	ADR-0009

Platform decisions that apply here and are not repeated: DuckDB in Lambda for serving, Airflow, Iceberg with Glue, k3s, Terraform, Parameter Store, UTC timestamps, dev/live separation. See ../Platform/adrs/.

Minor decisions folded from ADRs

These were separate ADRs and are recorded here in full because each is a setting, a wording rule, a format or a repeat of a platform rule. The old numbers are kept so references still resolve.

Old ADR	Decision	Detail
ADR-0005	The meteor shower calendar is stored as a small table of factual fields only	Only factual fields are stored. Decided by John, Oct 9, 2026: only the fields the score needs, each with its source: shower name, active dates, peak date and rate. Radiant position is stored only if the formula uses it; the proposed formula (ADR-0010) does not. The IMO is credited on the lookup page. No descriptive text, tables or images are copied from the site.
ADR-0011	Late or failed nightly data is shown with a late notice, never hidden	Decided by John, Oct 9, 2026: show the old data with a visible late notice and the time of the last good run. Do not hide it. For Project 4: when no row exists for tonight's `night_date` at the place, the API returns the most recent earlier row with `late` set to true and `last_good_run_time_utc`, and the page shows a visible late notice with that time in the visitor's local time. The publish gate still blocks bad data from being published (engineering standard 4); old published nights stay available. A place with no published row at all returns 404.
ADR-0012	VIIRS is kept as a manifest and re-downloaded once a year	Decided by John, Oct 9, 2026: treat VIIRS like the other unchanging archives. Store a manifest (URL, size, hash) and re-download the file once a year. The derived H3 resolution 8 table is stored in Iceberg.
ADR-0013	"Tonight" is the local night at the place, sunset to sunrise	Decided by John, Oct 9, 2026: "tonight" is the local night at the place, from sunset to sunrise in its time zone. The Moon is evaluated across that night. Times are stored in UTC and converted on the page [BP Engineering standard 15]. Proposed to implement it (not decided): hourly samples of the night; `night_date` is the local date of the sunset; sunset and sunrise from Skyfield; the cell's time zone from a time zone database. The time zone source and the sunrise and sunset method are OQ-03.
ADR-0015	The "clearly different" margin waits for the formula, and Phase C stays unspecified	The numeric margin for the "clearly different" check is set after John approves ADR-0010. Until then the check is written as the plan states it. Phase C stays unspecified in this document set. It needs its own design, and its done-when checks are written when that design is made. The gap is recorded here so it is not forgotten.
ADR-0016	The second ADR-0006 file is marked superseded	Keep the lower-48 file as the active ADR-0006. The other file was deleted on John's request, Oct 9, 2026, after a search found that no document links to it.
ADR-0017	"Tonight" before sunrise is the night still in progress	Between midnight and sunrise at the place, "tonight" is the night still in progress, the one whose sunset was the previous evening. From sunrise until the next sunset, "tonight" is the night that starts at the next sunset. The page shows the night's start and end in UTC, converted to local time on the page.
ADR-0018	The VIIRS file is downloaded by hand with a free EOG login	John downloads the annual VIIRS file by hand with his free EOG login. The pipeline reads the file that he downloaded. The manifest records the URL, size and hash, as ADR-0012 requires. There is no automated EOG download and no EOG credential in Parameter Store or in code.

15 Risks

Risk	Handling
The formula is approved (ADR-0010); the weights are starting values and are not validated	OQ-02 (approved); the 4B.3 backtest tunes the weights
"Clearly different" in the 4A.2 acceptance check is not measurable	OQ-13; the numeric margin is set after the formula is approved (decided by John, Oct 9, 2026)
A skipped or failed nightly run leaves tonight without a new score	ADR-0011: the page shows the latest earlier row with a late notice and the last good run time
VIIRS latitude coverage may stop short of Alaska or Hawaii or limit Phase B	OQ-01; Alaska and Hawaii wait for the first-run check; polar limits shown in Phase B
Data sizes unknown (VIIRS, GFS, resolution 8 cell count) so job limits cannot be set	OQ-05; measure first, as the build plan says [BP Data sizes: Measure first]
Cloud cover is about 28 km while light is much finer, so the score may look more precise than it is	ADR-0004; page wording; backtest
GFS file format or availability at 10:00 UTC differs from assumptions	First-run check; OQ-04

Risk	Handling
A source format changes	Contract diff before loading; weekly smoke test [BP Engineering standard 5, 1B.9]
IMO terms: only factual fields are allowed, and the calendar may not provide active dates	ADR-0005; OQ-14
Spot unavailable	Night skipped; reruns safe; the page shows the late notice (ADR-0011)
Vote abuse or personal data in votes	Follow Project 1's privacy rules (ADR-0009). The numbers are open. Do not build POST /votes until OQ-15 is answered
Phase C hourly updates conflict with the model where the session box is off most of the day	Phase C is unspecified and needs its own design (decided by John, Oct 9, 2026) [BP Engineering standard 16]
A migration of the platform serving template changes the API	Provisional API; align in OQ-19

16 Open questions

Each question has a matching GitHub issue (placeholder "issue: to be created"). Answered questions stay in this table, with the answer and its ADR number, so existing references still resolve. Each answer is also recorded in the section it affects. Open questions do not block coding of the parts they do not affect. Type: Spike = answered by running something; Decision = needs John's choice; Approval = John approves a proposal.

Answered by John on Oct 9, 2026 (now recorded in the sections and ADRs named)

Former question	Answer	Where recorded
What the page shows when tonight has no score	Show old data with a visible late notice and the last good run time; never hide it	ADR-0011, section 8, openapi.yaml
Phase A area (part of OQ-06)	Lower 48 first; Alaska and Hawaii once the VIIRS check shows latitude coverage	ADR-0006
Meaning of "tonight" (part of OQ-03)	Local night at the place, sunset to sunrise; Moon evaluated across the night; UTC stored	ADR-0013, 7.7.1
VIIRS raw storage (part of OQ-07)	Manifest (URL, size, hash) and a yearly re-download	ADR-0012
Vote privacy (part of OQ-15)	Follow Project 1: anonymous phone ID, limits per visitor and per day, no internet addresses	ADR-0009
Shower fields (part of OQ-14)	Only the fields the score needs, each with its source	ADR-0005
The "clearly different" check (part of OQ-13)	A numeric margin is set after the formula is approved	requirements.md F-A2
Phase C	Unspecified; needs its own design; done-when checks missing	ADR-0007, requirements.md
The score formula (OQ-02)	John approved the formula as written, Oct 9, 2026; the backtest tunes the weights	ADR-0010 (Accepted)
Tonight before sunrise (OQ-22)	Picked by Claude at John's request: before sunrise the night still in progress is shown	ADR-0017 (Default)
VIIRS download (OQ-23)	John chose a free login: the file is downloaded by hand; no automated download and no EOG credential	ADR-0018 (Accepted)

Decisions still needed from John

ID	What
OQ-02	Approved Oct 9, 2026 (ADR-0010). Not needed any more.
OQ-06	Score grain (H3 resolution), cloud-to-cell assignment, the boundary source for the lower 48, the "land cell" definition
OQ-03	Time zone source and the sunset and sunrise method (the night itself is decided)

ID	What
OQ-14	Whether the IMO calendar provides the active dates the formula needs
OQ-20	Forecast freshness threshold for the publish gate (and retries and timeouts)

Spikes that also gate Phase A work: OQ-01 (VIIRS latitudes, also decides Alaska and Hawaii), OQ-04 (GFS run, hours, format), OQ-05 (sizes and job limits), OQ-07 (VIIRS product details).

Open questions

ID	Question	Type	Blocks	What answering it needs	GitHub issue
OQ-01	What latitudes does the VIIRS annual file cover? (The plan's recorded open question for Project 4.) It decides when Alaska and Hawaii are added and the polar limits in Phase B.	Spike	4A.1 completion, Alaska and Hawaii, 4B.1	The file, downloaded with a free Earth Observation Group account	issue: to be created
OQ-02	**Answered (Oct 9, 2026): John approved the formula as written in ADR-0010. The backtest tunes the weights. Closed.** Is the proposed formula and its starting weights (ADR-0010) approved, or what changes? This includes the visitor-facing name "no active shower" (stored as no_active_shower) for the factor that replaces the plain name "shower". The weights are STARTING VALUES, not validated.	Approval	4A.2, 4B.2	John's approval	issue: to be created
OQ-03	**Answer 21 (Oct 9, 2026), ADR-0013: "tonight" is the local night, sunset to sunrise, with the Moon evaluated across it and times stored in UTC. Still open: the time zone source and the sunset method.** Which time zone source gives a cell's time zone, how are sunset and sunrise computed (Skyfield proposed), and how are GFS forecast hours mapped into the night window?	Decision	4A.2	John's choice; a first run	issue: to be created
OQ-04	Which GFS run is complete at 10:00 UTC, which forecast hours feed tonight, and what is the file format and reader?	Spike	4A.1	A first-run read of GFS files from AWS	issue: to be created
OQ-05	What are the sizes of the GFS run files and the VIIRS file, the number of cells and rows per night, and from them the maximum rows and runtime for each Spark job and the session box type needed?	Spike	job limits, 4A.2	Download and a test run on the session box	issue: to be created
OQ-06	**Answer 20 (Oct 9, 2026), ADR-0006: Phase A covers the lower 48 states first. Still open: the grain, cloud assignment, boundary source and land cell.** What H3 resolution is the score grain; how is a cloud grid point assigned to a resolution 4 cell; what source defines the lower 48 boundary; what defines a "land cell"?	Decision	4A.1, 4A.2, 4B.1	John's decision; boundary or land mask source	issue: to be created
OQ-07	**Answer 22 (Oct 9, 2026), ADR-0012: store a manifest and re-download once a year. Still open: product details and schedule. Automation is not used (OQ-23, answered); see OQ-24.** VIIRS product details: version and year, variable, unit, file format, pixel to cell aggregation; the exact schedule or trigger of the yearly re-download (the yearly re-download itself is decided, ADR-0012); how the download is automated and authenticated (credential in Parameter Store); how the weekly smoke test samples it	Spike	4A.1	The file and the account	issue: to be created
OQ-08	Which JPL ephemeris file does Skyfield use, from where, and under what terms?	Spike	4A.1	Skyfield and JPL documentation	issue: to be created
OQ-09	What tolerance counts as a match between Skyfield and Horizons for the 20 positions, which 20 positions, and are the Horizons reference values stored in the repository or queried live? (One test: the Phase B first-run check and the 4B.3 test.)	Decision and Spike	4B.3	John's tolerance; a Horizons query	issue: to be created

ID	Question	Type	Blocks	What answering it needs	GitHub issue
OQ-10	How does a typed place become a point: geocoding source and terms, whether the API receives a place string, coordinates or an H3 cell, and the privacy of lookup requests? (Check whether Project 1 or the serving template already defines this.)	Decision	4A.3	The serving template and Project 1 design	issue: to be created
OQ-11	**Answered (answer 1, Oct 9, 2026; ADR-0011): show the old data with a visible late notice and the time of the last good run; never hide it.** Question: a skipped night cannot be caught up, and there is no stale-data rule.	Decision	4A.3, QR-11	Answered; recorded in ADR-0011	closed
OQ-12	How does dbt on DuckDB read the Iceberg tables, and do Project 4's marts stay on DuckDB with no Snowflake or MetricFlow (the Technology table marks none)? (The Spark/dbt split is proposed in 7.6 to 7.8.)	Decision	4A.2	John's decision; platform dbt setup	issue: to be created
OQ-13	**Answer 25 (Oct 9, 2026), ADR-0015: the margin is set after ADR-0010 is approved. ADR-0010 is now approved, so the number is the next open item.** What numeric margin makes the full-Moon night and the shower-peak night "clearly different", and which two nights are fixtures? (Approach decided: set after the formula is approved.)	Decision	4A.2 acceptance	After OQ-02	issue: to be created
OQ-14	**Answer 24 (Oct 9, 2026), ADR-0005: only the fields the score needs, each with its source. Still open: the active dates and the meaning of the rate.** Does the IMO calendar provide the active dates, peak date and rate the formula needs; how is it retrieved (format, URL); what does the rate figure mean; how often is it updated?	Spike	4A.1, 4A.2	The IMO site and its terms	issue: to be created
OQ-15	**Answer 23 (Oct 9, 2026), ADR-0009: follow Project 1's privacy rules. Still open: the numbers and the rest of the list.** Vote numbers and details: limit numbers per visitor and per day, counter windows, the H3 resolution of a vote, retention or expiry, whether votes are copied nightly to the lake, whether point-in-time recovery stays on for this table (Project 1 turns it off for its hiker reports), and where counts are shown. (Privacy rules decided: follow Project 1.)	Decision	4B.3	John's decision	issue: to be created
OQ-16	Backtest method: how the score is compared with known peaks and the Project 2 fireball baseline, what "agrees" means, how the weights are tuned, and where the short note is published	Decision	4B.3	John's decision; Project 2 data	issue: to be created
OQ-17	Global Phase B: does GFS cover the globe in the AWS bucket, how are places without a sunset or sunrise on a date (polar) treated, and how does the page state the limits?	Spike and Decision	4B.1	OQ-01, OQ-03, OQ-06; a GFS read	issue: to be created
OQ-18	**Accepted in part (ADR-0014, Oct 9, 2026): each credit line uses the provider's own wording. VIIRS is not confirmed as public domain (EOG says many products are CC BY 4.0); the JPL licence is unverified.** Exact credit lines and terms notes for GFS, VIIRS, IMO, and the JPL ephemeris and Skyfield	Decision	4A.1 acceptance	Each source's terms	issue: to be created
OQ-19	API details: path names and response shapes aligned with the serving template, throttle and cache values, latency target for this endpoint, wording of the late notice with the platform, and how the lookup test with the session box off is automated	Decision	4A.3	The serving template	issue: to be created
OQ-20	Freshness targets (forecast freshness threshold in the publish gate; published data freshness), and Airflow retry counts and task timeouts	Decision	4A.2	John's decision	issue: to be created
OQ-21	Which usage events does the site emit, in the platform's event format (the platform defines the format and collection point; Project 2 builds the tables)?	Decision	4B.3	Platform and Project 2 design	issue: to be created

ID	Question	Type	Blocks	What answering it needs	GitHub issue
OQ-22	**Answered (Oct 9, 2026, by Claude at John's request, ADR-0017): before sunrise the lookup shows the night still in progress; from sunrise, the night that starts at the next sunset.** Which night does the lookup treat as "tonight" between midnight and sunrise (for example at 3 a.m. Arizona time, when the 10:00 UTC run has just started), and when does the run's row become visible? ADR-0013 defines the night by its sunset date but does not settle this.	Decision	4A.2, 4A.3	John's choice	issue: to be created
OQ-23	**Answered (John, Oct 9, 2026, ADR-0018): a free login; the VIIRS file is downloaded by hand; no automated download and no EOG credential.** EOG limits programmatic access via OpenID Client to paid subscribers from June 1, 2026 (EOG register page, read Oct 9, 2026). Is the VIIRS file downloaded by hand with a free login (no automation, no stored credential), or does John pay for automated access? This also decides whether the first-run check "VIIRS download works with a free EOG account" can pass as written.	Decision	4A.1, first-run check	John's decision on cost and terms	issue: to be created
OQ-24	**Partly answered (Oct 9, 2026, from a Google Earth Engine catalog page, a secondary source): it lists the V2.2 series through 2024. The EOG page gives 2012 to 2020 for V2 and no years for V2.2; no 2025 product was found. Confirm on the EOG page before the first run.** The EOG page does not state years for V2.2. Is there a newer annual product to load? If not, the yearly re-download in ADR-0012 has nothing new to fetch.	Spike	4A.1, ADR-0012	The EOG product list	issue: to be created

17 Glossary

Term	Meaning
Sky score	A number from 0 to 100 (proposed scale), computed by the documented formula for one cell and one night, that says how good the night is for seeing shooting stars
Top factor	The factor that takes the most points off the score: cloud cover, Moon, light pollution or no active shower
Cell	An H3 hexagon
H3	A hexagonal grid system; resolution 4 is used for GFS and resolution 8 for VIIRS [ADR-0004]
cell_to_parent	The H3 function that returns the larger cell containing a cell; joins across resolutions use it
GFS	NOAA Global Forecast System, run four times a day; here its total cloud cover
VIIRS	Visible Infrared Imaging Radiometer Suite, a satellite instrument; its annual nighttime lights composite measures light pollution
EOG	Earth Observation Group, the source of the VIIRS composite; free login needed, and programmatic access is paid-only from June 1, 2026 (OQ-23)
IMO	International Meteor Organization, source of the shower calendar
Shower peak	The date a meteor shower is strongest
Skyfield	A Python astronomy library used to compute the Moon's position
JPL ephemeris	A data file of planetary and lunar positions from NASA's Jet Propulsion Laboratory
Horizons	JPL's online service for positions; used here only as a test reference
Moon illumination	The percentage of the Moon's visible disc that is lit
Moon altitude	The Moon's angle above the horizon
Fireball baseline	Project 2's fireball data used as a comparison in the backtest
Nightly run	The scheduled pipeline run on the session box at 10:00 UTC

Term	Meaning
Session box	The temporary AWS computer (a spot instance) that runs k3s, Airflow and Spark
Spot instance	A cheaper AWS computer that AWS can take back with a two-minute warning
k3s	A lightweight Kubernetes
Iceberg	A table format on S3, registered in the Glue catalog
Parquet	A columnar file format used for published tables
DuckDB	The embedded query engine used in Lambda
dbt	Tool that builds layered data models with tests
Great Expectations	Tool for data quality checks; used here as the publish gate
Write, audit, publish	New data is first stored in staging, passes checks, then goes live
Idempotent	Safe to run again with the same result
ODCS	Open Data Contract Standard, the format of the data contracts
OpenAPI	The format of the API specification
ADR	Architecture decision record
C4	A model for architecture diagrams: context, containers, components
Open question	A question only a spike or a decision can answer, each with a GitHub issue
PITR	Point-in-time recovery for DynamoDB
Parameter Store	AWS Systems Manager service holding every key
Cold start	The delay on the first request after a period of inactivity
Night	The local night at a place, from sunset to sunrise in its time zone (ADR-0013)
DAG	A directed acyclic graph: Airflow's name for one scheduled pipeline made of ordered tasks
Partition	One slice of a table, here the rows of one night; a rerun replaces the whole slice
Manifest	A table that lists downloaded files with URL, size, fetch time and hash instead of keeping the files
Hash	A fingerprint of a file's contents that changes if the contents change
Mart	A final table built for use, such as the published score table
Gold table	A final, checked table that is safe to publish
Lambda	The AWS service that runs the lookup code without a server of our own
UDF	A user-defined function that Spark calls on each row
Smoke test	A quick check that a source still returns data in the expected format
Spike	A short test that answers an open question by running something
Engineering standard	One of the 16 numbered rules in the build plan's Engineering standards table (for example standard 3: tasks overwrite a partition and never append blindly)
Kafka	A message queue for live data; not used in Project 4
Snowflake	A cloud data warehouse; not used in Project 4
MetricFlow	A tool that defines business metrics; not used in Project 4
Superset	A dashboard tool for warehouse data; not used in Project 4
Grafana	A dashboard tool for system health; not used in Project 4
OpenLineage	A standard for records of which data each job read and wrote
Terraform	A tool that creates cloud resources from code
OIDC	OpenID Connect; lets GitHub Actions get short-lived AWS access without stored keys
FastAPI	A Python web framework used for the lookup endpoint

Term	Meaning
API Gateway	The AWS service that exposes the Lambda function over HTTP
CloudFront	The AWS service that serves the site's files
X-Ray	The AWS service that records traces of requests
Object Lock	An S3 setting that blocks deletion or overwrite of a file for a set time
SCD Type 2	Slowly changing dimension: a way to keep old versions of changing rows
LLM judge	A language model that scores another model's answers; not used in Project 4
Landing (input) table	The table where an input is first stored; the build plan says an input "lands" there
Late notice	The visible message shown with old data when tonight's score is missing, with the time of the last good run (ADR-0011)
Points lost	How many of 100 points each factor takes away from the score (ADR-0010)

Part 4: Operations

Shooting Star Forecast/operations.md

Project 4 operations

Field	Value
Status	Draft for review. Written before coding, from the design doc, the ADRs and the contracts. Measured times and costs replace the estimates here when the first runs exist
Owner	John (GitHub: jwasikye)
Reviewers	John approves this document before Phase A coding starts
Last updated	Oct 9, 2026
Related	design-doc.md (sections 8, 11, 12 and 13), security-and-privacy.md, ../Platform/operations.md, ../Platform/platform-requirements.md

This document says how Project 4 runs day to day: when things happen, what is watched, what a person does when something breaks, and what the project costs. Every value is either a recorded decision or a confirmed fact. Where a value is not decided yet, the row names the open question (OQ-xx) that decides it.

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1. Schedule

Job	When	Where it runs	Maximum run time
Nightly run: ingest, shower calendar, Moon, conditions, score, dbt, publish gate, metrics, Iceberg maintenance last	Every night at 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern)	Session box on k3s, spot only, shut down when the pipelines finish	2 hours. About 45 minutes is expected (QR-03)
VIIRS annual load (ADR-0012)	Once a year. John downloads the annual file by hand with a free EOG login (ADR-0018). The exact trigger is open (OQ-07)	Session box	Not recorded (OQ-05)
Weekly source smoke test	Once a week, as platform step 1B.9	Scheduled	Not recorded. How it samples VIIRS is open (OQ-07)
Lookup endpoint (Phase A)	On each request	Lambda behind API Gateway. Works with the session box off	The first request after inactivity may take up to 10 seconds (design 8.2)
Vote endpoint (Phase B)	On each vote	Lambda, writing to DynamoDB	Not recorded

Backfills run only during work sessions, not in the nightly window (platform decision).

2. Freshness

- Every night the pipeline should publish tonight's row for each published place (design 8.1).
- The freshness threshold that the publish gate uses is not decided (OQ-20). The contract for the score table reads "undecided, see OQ-20".
- Each run writes its freshness and row counts to the metrics table, and the status page shows the last good run (design 12.3, QR-13).
- When tonight's row is missing, the lookup returns the most recent earlier row with `late` set to true and the time of the last good run. The page shows a late notice with that time in the visitor's local time (ADR-0011).
- A skipped or failed night cannot be caught up later. Nothing rebuilds it (ADR-0011).

3. Monitoring and alerts

What is watched	Where it shows	Who is told
Freshness and row counts for each run	Metrics table; status page	John, through the status page
A failed run or a missed freshness target	Email. One de-duplicated GitHub issue per failure	John
A real incident	One incident record	John
A job over its maximum rows or runtime	Stops before publishing; opens an incident (design 8.1, step 5)	John, through the incident
The nightly run reaching its 2-hour maximum	Incident (design 8.1, step 10)	John, through the incident
Lookup errors (500) and throttling (429)	Page shows an error message. Throttle values are open (OQ-19)	Visitor sees the message; John checks the logs
Source format changes	Schema diff before loading; weekly smoke test (QR-14)	John, through the failed task or the weekly result

Phase B adds OpenLineage and the dbt docs catalog (design 12.3).

4. What happens when something fails

Failure	What the system does	What a visitor sees	What John does
No spot capacity at 10:00 UTC	The night is skipped. No new score exists for it	The latest earlier row, with the late notice and the last good run time	Nothing at once. The next night's run starts as usual. The skipped night is not recovered (ADR-0011)
GFS file missing or unreadable	The ingest task fails, which opens an incident and sends an email (design 8.1, step 3)	The late notice	Read the incident. Rerun the nightly run once the source is back
Source format differs from its contract	The task fails before any write (design 8.1, step 4)	The late notice	Check the source change and decide the fix before rerunning
A job goes over its maximum rows or runtime	The job stops before publishing and opens an incident	The late notice	Read the measured size. The limit values are open (OQ-05)
A dbt test fails	The task fails. Nothing is published	The late notice	Fix the cause and rerun
Publish gate fails (freshness or a value range)	Nothing is published. Earlier published nights stay as they are. An incident is opened and an email is sent (design 8.1, step 7)	The latest earlier row, with the late notice	Read which check failed. Rerun after the fix
Nightly run reaches 2 hours	The session box shuts down. An incident is recorded (design 8.1, step 10)	The late notice if tonight's row is missing	Read the metrics to find the slow task
AWS reclaims the spot instance	The platform's interruption handler reacts to the two-minute warning. Every task is safe to rerun	As above	Rerun the nightly run if it did not finish
Lookup Lambda error	The API returns 500	An error message on the page	Check the Lambda errors
Place with no published data at all	The API returns 404 with a reason	A "no data for this place" message	Nothing

5. Runbooks for real failures

No real incident has been recorded for this project yet. The steps below come from the design. After the first real incident, check them against what happened, and record the blameless postmortem with the incident.

A. Nightly run late or failed

1. Open the incident and the run's row in the metrics table.
2. Find the task that failed: ingest, schema check, job limit, dbt test or publish gate.
3. Fix the cause. If the source is down, wait for it to return.
4. Rerun the nightly run. Rerunning is safe: a night's partition is overwritten only after the run succeeds, so a repeat gives the same rows (QR-09).
5. Check the status page. The last good run time should update, and tonight's late notice should clear.

B. Publish gate blocked a night

1. Read which check failed: freshness or value range.
2. Do not force the data through. Earlier published nights keep being served with the late notice.
3. Find the cause in the source data or the job, fix it, and rerun.

C. Nightly run hit the 2-hour maximum

1. Open the incident and find the longest task in the metrics.
2. Compare the row counts with the measured size estimate. If they grew, the job limit may need a decision (OQ-05).
3. Check whether the night's score was published. It is published only if the publish gate passed.

D. Lookup returns 500 or 429

1. For 500, check the Lambda errors for the time of the visitor's request.
2. For 429, the API is throttling. Check the traffic. The throttle values are open (OQ-19).

E. VIIRS annual file

1. Download the file by hand with the free EOG login (ADR-0018).
2. Record its URL, size and hash in the manifest (ADR-0012).
3. Check its latitude coverage (OQ-01).
4. If the download or load fails, the previous year's table stays in use (design 8.4).

6. Keys and access

- Every key is in AWS Parameter Store. None is in code or in GitHub (QR-12).
- The VIIRS file is downloaded by hand. No EOG credential is stored anywhere (ADR-0018).
- Each pipeline has one least-privilege IAM role. Dev roles cannot write live data (design 12.1).
- Deploys use GitHub Actions with OIDC. Tests run on sample data first, then dev, then live (design 8.5).

7. Cost

Item	Limit or fact	Source
Whole platform budget	Under $50 a month, browser only	Design 3.3
AWS budget backup	$40	Design 12.5
Session limit	4 hours, with idle shutdown as the main guard	Design 12.5
Nightly run	2 hours maximum	Design 12.5; ADR-0012
Each Spark job	A maximum number of rows read and written, and a maximum runtime. The values are open	OQ-05
Model API	None. The project makes no model calls	ADR-0001
Votes (Phase B)	DynamoDB with provisioned capacity inside the free tier. Heavy traffic is throttled, not billed	ADR-0009
Measured cost per run	Not recorded yet. Measured after the first runs	OQ-05

8. Reruns and backfills

- Rerunning a night overwrites only that night's partition, and only after success (design 8.1, step 7; QR-09).
- The Iceberg maintenance task is the last task of the nightly run, so it never competes with the pipelines (design 8.1, step 9).
- Backfills run only during work sessions, because they would not fit in the nightly window (platform decision).

9. Open items that affect operations

ID	Question	Why it matters here
OQ-04	Which GFS run is complete at 10:00 UTC, and which forecast hours feed tonight	Decides when the nightly run can read its input
OQ-05	Sizes of the GFS and VIIRS files, and the job limits	Sets the maximum rows and runtime for each job, and the cost per run
OQ-07	The VIIRS product details, the exact trigger of the yearly load, and how the weekly test samples it	Decides the annual load and the smoke test
OQ-19	API throttle and cache values, and whether the lookup test with the box off is automated	Decides the 429 behavior and the end-to-end check
OQ-20	Freshness threshold for the publish gate, retry counts and task timeouts	Decides when a missed night counts as late

Part 5: Security and privacy

Shooting Star Forecast/security-and-privacy.md

Project 4 security and privacy

Field	Value
Status	Draft for review. Written before coding, from the design doc, the ADRs and the contracts. The threat model is extended when the Phase B design is settled
Owner	John (GitHub: jwasikye)
Reviewers	John approves this document before Phase A coding starts
Last updated	Oct 9, 2026
Related	design-doc.md (sections 3.3, 12.1, 12.2 and 12.4), requirements.md (QR-07, QR-08, QR-12), operations.md, ../Platform/security-and-privacy.md

This document covers what could go wrong with Project 4 and how it is prevented, what personal data the project holds, how long it is kept, and which licences and credit lines apply to its inputs. Every value is either a recorded decision or a confirmed fact. Undecided values name their open question (OQ-xx).

The platform's own rules for keys, storage and access are in ../Platform/security-and-privacy.md. They apply to this project and are not repeated in full here.

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1. What the project holds

Asset	Phase	Why it matters
Published forecast table (score, top factor, late flag, night times)	A	Shown to every visitor. It must be correct and must not be published from bad data
Source inputs (GFS, VIIRS manifest, IMO calendar, JPL ephemeris, Skyfield)	A	Each has a terms note and a credit line
Keys and secrets	A and B	Any leak gives access to AWS. Keys are in Parameter Store only
Vote records (cell, answer, time)	B	Personal data risk if linked to a visitor. See section 3
Visitor limit counters	B	Stored apart from votes and expire. See section 3

2. Threat model

#	Threat	What could happen	Control	Status
T1	A key or password is written into code, a repository or a backup	An attacker gets access to AWS or to a source account	Every key is in Parameter Store (SecureString). None in code or in GitHub. gitleaks (secret scanner) runs in CI (QR-12; engineering standard 13)	In the design. CI check to be built
T2	A pipeline role is compromised or misused	Data is written where it should not be	One least-privilege IAM role per pipeline, read-only where possible. Dev roles cannot write live data (design 12.1)	In the design. The dev-cannot-write-live check is a platform check
T3	A source page contains instructions (for example the IMO page)	The pipeline acts on text it was meant only to read	Source text is data, never instructions. Only the factual fields are kept (design 12.1; ADR-0005)	In the design
T4	A container image has a known vulnerability	The image could be used to attack the system	Trivy (image scanner) runs in CI (QR-12)	In the design. CI check to be built
T5	Bad or changed source data reaches the site	Wrong or misleading scores are shown	Schema diff against each contract before loading. dbt tests. Publish gate checks freshness and value ranges. Nothing is published when a check fails (QR-10, QR-14)	In the design. CI tests to be built

#	Threat	What could happen	Control	Status
T6	Votes are sent in bulk to skew the counts (Phase B)	Vote counts are wrong. Cost rises if traffic is billed	Limits per visitor and per day. Counters stored apart from votes and expiring. Heavy traffic is throttled, not billed (ADR-0009)	The numbers are open (OQ-15)
T7	A vote is linked to a person (Phase B)	Personal data is exposed	The anonymous random phone ID stays on the phone. No internet addresses. The phone converts the position to an H3 cell, so exact coordinates never reach the server (ADR-0009)	Rules decided. Retention open (OQ-15)
T8	The lookup API is flooded	Slow or failed lookups; cost	API Gateway throttling returns 429 (design 8.2)	Throttle values open (OQ-19)
T9	Lookup coordinates reveal where a visitor is	Privacy loss	Open. Whether coordinates are sent, and whether they are logged, is decided with the lookup design	OQ-10
T10	A source is used without its licence or credit	Breach of its terms	Terms note and credit line for each input, shown on the page (QR-08; ADR-0014)	Two inputs are unverified (section 6)

3. Personal data

No personal data is intended for Phase A. The contract for the votes table says "none intended", to be confirmed in this document. That confirmation depends on two open questions: lookup privacy (OQ-10) and vote retention (OQ-15). Until both are answered, the statement below is a plan, not a finished check.

Data	Where it is	Why it is kept	Who can read it	How long it is kept	Status
Lookup coordinates or place (Phase A)	Sent from the visitor's browser to the lookup API	To find the cell for the place	The lookup Lambda	Not decided (OQ-10)	Open
Usage events from the site	Platform event format	Measure use	Not recorded for this project	Not recorded for this project	No raw IP address in any usage event (QR-07; BP 2B.4)
Anonymous random phone ID (Phase B)	Phone only	Count limits per visitor	The visitor's phone	On the phone	Decided (ADR-0009)
Vote (Phase B): H3 cell, answer, time in UTC	DynamoDB	Show that people saw a shower	Not decided (OQ-15)	Not decided (OQ-15)	Open. The record carries no ID (design 8.3)
Visitor limit counters (Phase B)	DynamoDB, apart from votes	Enforce the per-visitor and per-day limits	The vote Lambda	Expire. The time is open (OQ-15)	Open

Point-in-time recovery (restoring a table to an earlier time) is on for the platform's DynamoDB tables. Whether it stays on for the votes table is open (OQ-15). Project 1 turned it off for its hiker reports, and that decision covers that table only (ADR-0009).

4. Privacy rules that apply

- Every stored timestamp is in UTC. Only the page converts to local time (design 12.7; engineering standard 15).
- No raw IP address is stored in usage events (QR-07).
- The phone turns the position into an H3 cell before anything is sent. Exact coordinates never reach the vote endpoint (ADR-0009).
- Votes follow Project 1's rules: an anonymous ID kept on the phone, limits per visitor and per day, and no internet addresses (decided by John, Oct 9, 2026).

5. Deletion periods

Project 4 uses the platform's storage clean-up rules (platform decision, Oct 9, 2026). They apply to this project's buckets:

Data	Kept for
Dev data	14 days
Temporary files	7 days
Failed uploads	7 days

Data	Kept for
Pipeline logs	90 days
Security logs	1 year
Old object versions	30 days
Raw and published data	Kept. Live raw files are locked against deletion or overwrite for 30 days

The vote and counter periods are not decided (OQ-15). Those are listed in section 3.

6. Licences and credit register

Each input's terms are checked on its provider's own page (ADR-0014). The project is a non-commercial portfolio. Check licensing again before any charging (design 12.4).

Input	Provider	Terms as recorded	Credit	Status
GFS cloud cover	NOAA, through the AWS open data bucket	NOAA asks for attribution	Wording open (OQ-18)	Attribution required. Exact wording open
VIIRS nighttime lights	Earth Observation Group	EOG's page says many products are CC BY 4.0. The annual composite's licence is not confirmed. The public-domain claim in the build plan is not confirmed	Wording open (OQ-18)	Annual composite UNVERIFIED
IMO shower calendar	International Meteor Organization	Copyright notice with no reuse terms. Only factual fields are kept (ADR-0005)	Credit line required	Limited to factual fields
JPL ephemeris	NASA Jet Propulsion Laboratory, Solar System Dynamics	The SSD page has no licence statement. Cite Park et al. 2021 (ADR-0014)	Citation	UNVERIFIED
Skyfield (library)	Open-source project	MIT licence (ADR-0014)	MIT notice	Confirmed

Each input also has a terms note, and its credit line is shown on the lookup page (QR-08; `source_credits` contract).

7. Access and change control

- Keys are read at run time from Parameter Store. The VIIRS file is downloaded by hand, so no EOG credential exists in the system (ADR-0018).
- Dev and live have separate storage, catalogs and roles. Dev cannot write live data (design 12.1).
- Only merges to the main branch reach live, and only after the changed pipelines pass on sample data (design 8.5).

8. Open items

ID	Question	Why it matters for security or privacy
OQ-10	How a typed place becomes a point, the geocoding source and terms, and the privacy of lookup requests	Decides T9 and whether the lookup sends or logs coordinates
OQ-15	Vote limits, counter windows, retention and expiry, copy into the lake, and point-in-time recovery	Decides T6, T7 and section 3
OQ-18	Exact credit lines and terms notes for GFS, VIIRS, IMO, the JPL ephemeris and Skyfield	Decides T10 and the credit lines on the page
OQ-19	API throttle values	Decides T8
OQ-07	How the weekly smoke test samples VIIRS without a stored login	Decides whether the weekly test needs any credential

Part 6: Architecture decision records

Shooting Star Forecast/adrs/0001-sky-score-is-a-formula-not-a-model.md

0001. The sky score is a documented formula, not a model

- Status: Accepted (decided by John, Oct 9, 2026)
- Project: 4, Shooting star forecast
- Related: ADR-0002

Context

Project 4 gives tonight's chance of seeing a shooting star at a location and the top factor behind it. Its four inputs are cloud cover (from NOAA's Global Forecast System, GFS), the Moon (computed with the Skyfield library), light pollution (from VIIRS, a satellite record of nighttime lights) and the meteor shower calendar (from the International Meteor Organization, IMO) [BP 4A.1].

The build plan says AI stays minimal on purpose: the score is a documented formula, and a decision record says why a model would not improve it [BP Project 4]. John's recorded decision: "AI use: Minimal on purpose; the score is a formula." Reason: "Not every project needs a model." [BD Projects 3, 5, 2 and 4, Project 4 row].

The Technology by project table lists no machine learning, MCP server, agent, LLM evaluation or LLM judge for Project 4. The Documentation Plan lists no model card for Project 4 ("No (no model)").

Decision

The score is a formula over the per-cell, per-night conditions. The formula names the top factor (the input that lowers the score the most) for every score, and its weights are documented [BP 4A.2]. No trained model and no language model call is used in the pipeline, the API or the page.

The formula itself is in ADR-0010, which John approved on Oct 9, 2026, as written. Step 4A.2 may be built.

Alternatives considered

1. A trained model that predicts visibility. John did not choose this. His reason is "Not every project needs a model." Reasoning that follows from the plan: the only outcome data the plan creates is the "did you see one?" votes, which start in Phase B (4B.3), so there are no outcome labels while the score is first built in Phase A; and step 4A.2 requires the top factor to be named, which a formula gives directly.
2. Model-generated explanation text. Not chosen; see ADR-0002.

No measured comparison of a model against the formula exists, and none is planned in the build plan. The Phase B backtest (4B.3) compares the formula's scores with known shower peaks and the fireball baseline, which shows where the formula is weak but does not train anything.

Consequences

- No model card is written for this project [DP].
- No model API spend for this project, no eval harness, no LLM judge.
- The formula and weights are specified in ADR-0010 and must be approved and version-controlled before 4A.2. Any change to the formula changes a documented score, and the project records it in the changelog.
- Votes (4B.3) are stored and counted but do not feed any model. Using them as labels later would need a new ADR.

0002. Explanations are pre-written phrases filled with computed values

- Status: Accepted (follows build plan step 4B.2 and ADR-0001; recorded Oct 9, 2026)
- Project: 4, Shooting star forecast
- Related: ADR-0001

Context

Step 4B.2 specifies "Pre-written phrases for each top factor and combination (for example, the Moon is 92% lit and drowns out the shower), filled with computed values." Its done-when check is "Every phrase's values match the score table." [BP 4B.2]

AI is minimal by design in this project (ADR-0001). Project 3's buoy fault flags likewise show "no generated note" [BP 3B.6], and Project 2's brief checks every model-written number against the tables before publishing [BP 2A.3]; Project 4 needs no such check because no text is generated.

Decision

Explanation text is a fixed set of phrase templates stored in the repository, one for each top factor and each recorded combination of factors. A template contains named slots. The slots are filled with values read from the score table row for the looked-up cell and night. No model writes or rewrites any explanation text.

Alternatives considered

1. Model-generated explanations. Not chosen. The plan specifies pre-written phrases, and numbers in generated text would need a separate verification step.

No other alternatives are recorded in the sources.

Consequences

- Every top factor and every combination the formula can produce needs a phrase. The set of combinations depends on the formula once approved (ADR-0010); the formula stores every value a phrase needs.
- An automated test renders phrases from the score table and compares each embedded value with the table (acceptance criterion for F-B2).
- Changing wording is a code change reviewed in a pull request.
- Phrase values come from the same row as the score, so a phrase cannot disagree with the score shown.

0003. Moon altitude and illumination come from Skyfield with a JPL ephemeris

- Status: Accepted (build plan step 4A.1; recorded Oct 9, 2026)
- Project: 4, Shooting star forecast

Context

The score needs the Moon's altitude (its angle above the horizon) and illumination (the lit share of its disc) for each cell and night. Step 4A.1 says these are "computed with Skyfield and a JPL ephemeris" [BP 4A.1].

Confirmed fact: Skyfield uses JPL ephemerides (data files of planetary and lunar positions from NASA's Jet Propulsion Laboratory) and agrees with the U.S. Naval Observatory to within 0.00001 arcseconds [DP Answered by research: Moon calculation accuracy; BP Moon calculations]. The build plan keeps a separate check: 20 Moon positions from

Skyfield are compared with JPL Horizons (NASA's online positions service) within a stated tolerance. The plan lists this check as a Phase B first-run check and again in step 4B.3 [BP Project 4 Phase B; BP Risks table].

Decision

Moon altitude and illumination are computed inside the pipeline with Skyfield and a JPL ephemeris file. The project uses JPL Horizons only as an independent reference in a test, never as a data source at run time.

Which ephemeris file is used, its terms and credit wording are open (OQ-08, OQ-18). The test tolerance is open (OQ-09).

Alternatives considered

The build plan and decisions record no alternative Moon calculation. Calling Horizons at run time is not planned.

Consequences

- No external service call is needed for the Moon at run time; the pipeline needs the ephemeris file as an input (contract: jpl_ephemeris).
- The Moon calculation always gives the same result for the same cell location and time, so a unit test can check it with fixed inputs.
- The lookup page must credit the ephemeris file.
- The 20-position Horizons test must run in CI (Phase B) and needs stored reference values or another way to obtain them (OQ-09).

`Shooting Star Forecast/adrs/0004-h3-resolutions-and-cell-to-parent-joins.md`

0004. GFS cloud cover at H3 resolution 4, VIIRS at resolution 8, joined with cell_to_parent

- Status: Accepted (build plan step 4A.1 and engineering standard 9; recorded Oct 9, 2026)
- Project: 4, Shooting star forecast

Context

Two of the inputs use very different grids. GFS total cloud cover is a 0.25 degree grid, about 28 km cells [DP Answered by research: Cloud cover data]. The VIIRS annual composite is a much finer raster (a grid of pixel values). H3 divides the Earth into hexagonal cells, and its resolution number sets the cell size. Step 4A.1 assigns H3 resolution 4 to GFS and H3 resolution 8 to VIIRS [BP 4A.1].

Engineering standard 9 says to pick the H3 resolution from the source (the cell just larger than the source pixel) and to join across sources with cell_to_parent (the H3 function that returns the larger cell containing a cell). It names resolution 4 for 0.25 degree weather grids and resolution 8 for 15 arc-second rasters. The recorded reason is "Avoids false precision and mismatched joins." [BP Engineering standard 9]

Decision

- GFS cloud cover is stored at H3 resolution 4.
- VIIRS light values are stored at H3 resolution 8.
- A resolution 8 cell reads its cloud cover from its resolution 4 parent, using cell_to_parent.

The H3 resolution of the published score table (the score grain) is not recorded in the sources and is open question OQ-06.

Alternatives considered

The sources record the rule and its reason but no alternative resolution scheme for Project 4.

Consequences

- All resolution 8 cells inside one resolution 4 cell share the same cloud cover value. The page wording must not suggest finer cloud detail than the source has (see design doc section 12).
- The pipeline computes the Moon from a cell's location, so the Moon values have the resolution of the score grain (OQ-06).
- Joins between sources use cell_to_parent, so the run needs no point-in-polygon tests (checks of which area contains a point).
- The number of resolution 8 cells for the lower 48 states and later the world drives table size and the Spark job limits (OQ-05).

`Shooting Star Forecast/adrs/0006-lower-48-first-then-viirs-coverage.md`

0006. Phase A covers the lower 48 states, Alaska and Hawaii follow the VIIRS check, Phase B covers every land cell the VIIRS data covers

- Status: Accepted (build plan phasing; recorded Oct 9, 2026)
- Project: 4, Shooting star forecast
- Related: ADR-0004

Context

Each project is built in phases; Phase A is "the smallest version that works end to end, on a small area or sample, using the same stack as the full version" [BP How the plan works]. The build plan sets Project 4 Phase A scope to the United States [BP Project 4]; John narrowed it on Oct 9, 2026 (see Decision). Step 4B.1 extends to "every land cell the VIIRS data covers" and states the latitude limits on the page [BP 4B.1].

The latitudes the VIIRS annual file covers are not yet known. The Documentation Plan lists this as an open question. Answering it needs the file, downloaded with a free Earth Observation Group (EOG) account [DP Questions still open, Project 4].

Decision

- Phase A answers lookups in the lower 48 states first (decided by John, Oct 9, 2026). Alaska and Hawaii are added once the VIIRS first-run check shows its latitude coverage (decided by John, Oct 9, 2026).
- Phase B answers lookups for every land cell the VIIRS data covers, anywhere in the world, and shows the latitude limits on the page.

Alternatives considered

Global coverage from the first release is not chosen, because Phase A is meant to be the smallest end-to-end version. No other alternative is recorded.

Consequences

- The boundary source for the lower 48 is not recorded (OQ-06). If the VIIRS file does not cover some latitudes, Alaska and Hawaii wait for OQ-01.
- The term "land cell" needs a definition (OQ-06, OQ-17).
- Lookups outside coverage return 404 with reason outside coverage (design doc section 10).
- The score formula and pipeline must not hard-code the lower 48, so that Phase B only extends the data and needs no rewrite.

0007. Tonight's score is computed in the nightly run from that morning's GFS forecast

- Status: Accepted (John's defaults and decisions on the nightly run; recorded Oct 9, 2026)
- Project: 4, Shooting star forecast
- Related: ADR-0008

Context

Decisions recorded by John:

- Data freshness: a nightly run every day, not weekly. Reason: "Only about $1 a month more than weekly, keeps everything fresh, one system instead of two." [BD Data freshness]
- Nightly run time default: 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern). Reason: Project 2's asteroid brief is ready by morning, "tonight's sky score uses that morning's forecast", and Project 1's satellite windows cover a full hiking day. [BD Defaults used]
- The nightly run uses spot instances only (cheaper AWS servers that AWS can take back at short notice). When none is available, the night is skipped and the next run catches up safely [BP How the box is run]. The hard maximum is 2 hours; about 45 minutes is expected [BD Nightly shutdown].

GFS runs four times a day and publishes hourly total cloud cover out to 120 hours [DP Answered by research: Cloud cover data]. Phase C lists "hourly forecast updates" [BP Phase C].

Decision

Tonight's score for every cell is computed once a day by the nightly pipeline from that morning's GFS forecast and published as Parquet. The lookup reads the published file; nothing is computed when a visitor asks.

The sources do not record which GFS run is complete by 10:00 UTC or which forecast hours feed "tonight" (OQ-04). "Tonight" is the local night at the place (ADR-0013).

Alternatives considered

1. Weekly refresh. Not chosen; see the freshness decision above.
2. Hourly updates. Deferred to Phase C by the build plan.

No other alternative is recorded in the sources.

Consequences

- The page works with the session box off, because it reads precomputed Parquet files through Lambda and DuckDB.
- If a nightly run is skipped or fails, no new score exists for that night. Decided by John, Oct 9, 2026: the page shows the old data with a visible late notice and the time of the last good run (ADR-0011).
- Phase C hourly updates conflict with the model where the session box is off most of the day; anything that runs around the clock writes only to always-on storage, standard 16. Decided by John, Oct 9, 2026: Phase C stays unspecified and needs its own design; its done-when checks are missing and are written when Phase C is designed.
- The publish gate (freshness and range checks) runs inside the nightly run's 2-hour limit.

0008. The lookup page reuses Project 1's map and the shared serving template

- Status: Accepted (build plan step 4A.3 and the tools table; recorded Oct 9, 2026)

- Project: 4, Shooting star forecast

Context

Step 4A.3 says "Reuse Project 1's map: enter a place and see tonight's score and top factor, with every source credited." [BP 4A.3] The build plan's tools table fixes one stack for all apps and APIs: React and TypeScript, FastAPI on Lambda behind API Gateway (HTTP type for public pages). Reason: "One template reused by every project" [BP Tools table: Apps and APIs]. The serving template (the platform's shared starting point for sites and APIs) provides a map component, a lookup form, an answer card and a status page, and a publish step that writes gold tables as Parquet for DuckDB in Lambda [BP 0.4].

Project 1 is built before Project 4 [BP How the plan works: order].

Decision

The Project 4 page and endpoint are built on the serving template and Project 1's map component. Project 4 adds its own answer card content (score, top factor, credits), its own DuckDB query over its own published Parquet, and in Phase B its explanation text, latitude notice and vote button.

Platform and Project 1 documents are linked, not repeated: ../../Platform/design-doc.md, ../../Will I have signal out there/design-doc.md.

Alternatives considered

A separate map implementation for Project 4 is not chosen; the plan reuses one template. No other alternative is recorded.

Consequences

- Project 4's page depends on the template's component interfaces. A breaking change in the template requires a matching change in Project 4.
- The cold-start wording and measurement come from the template and the platform operations step.
- The template or Project 1 may define how a typed place becomes coordinates or a cell, but the sources do not record it (OQ-10).
- Preview deploys for every pull request come with the template [BP 0.4].

`Shooting Star Forecast/adrs/0009-votes-stored-in-dynamodb.md`

0009. "Did you see one?" votes are stored in DynamoDB

- Status: Accepted (build plan step 4B.3 and the visitor data decision; recorded Oct 9, 2026)
- Project: 4, Shooting star forecast (Phase B)

Context

Step 4B.3 adds a "did you see one?" button that stores votes in DynamoDB [BP 4B.3].

John's visitor data decision: DynamoDB. Reason: "Earlier options did not handle visitors writing at any hour; DynamoDB is always on and free at this traffic." The recorded platform decision also turns on point-in-time recovery (restoring a table to an earlier moment): "A bad write or deletion can be undone, for pennies." [BD Tools and platform: Visitor data; BD Data storage: DynamoDB]. The alternative the plan lists as removed for this job is PostgreSQL on the session box, which is off most of the time [BP Tools table].

Hard limit on DynamoDB: provisioned within the free tier, so heavy traffic is throttled rather than billed [BP Hard limits table]. Anything that runs around the clock writes only to always-on storage [BP Engineering standard 16].

Decision

Votes are written by the Lambda behind API Gateway to a DynamoDB table with provisioned capacity within the free tier.

Privacy and abuse rules (decided by John, Oct 9, 2026): follow Project 1: an anonymous random phone ID kept on the phone only, limits per visitor and per day, and no internet addresses. As in Project 1 step 1B.8, the phone converts its position to an H3 cell so exact coordinates never reach the server, and the limit counters are stored apart from the votes and expire, so a vote never carries an ID that links it to a visitor. The limit numbers stay open (OQ-15), as do the vote fields beyond those in the contract, the H3 resolution of a vote, the retention period, the copy into the lake, and where counts are shown. Whether point-in-time recovery stays on for this table is also open: John turned it off for Project 1's hiker reports table so that deletion is final, and that decision covers that table only.

Alternatives considered

PostgreSQL on the session box. Not chosen, because the session box is off most of the time. No other alternative is recorded.

Consequences

- Visitors can send votes at any hour, even with the session box off.
- Heavy traffic is throttled rather than billed.
- The table is visitor data, so the contract (contracts/feedback_votes.yaml) and the security and privacy document (written while building) must state what is stored and for how long.
- A nightly copy into the lake is the platform pattern for visitor data [BP Technology introduced table: DynamoDB visitor data]. Whether Project 4 uses it is OQ-15.

`Shooting Star Forecast/adrs/0010-sky-score-formula.md`

0010. Sky score formula and starting weights

- Status: Accepted (approved by John on Oct 9, 2026, as written). The weights are starting values for the 4B.3 backtest to tune.
- Project: 4, Shooting star forecast
- Related: ADR-0001, ADR-0002, ADR-0013

Context

Decided by John, Oct 9, 2026: Claude proposes the ingredients and starting weights in an ADR for John's approval; the 4B.3 backtest tunes them; nothing is built until approved. John wants to approve the numbers himself and have a measured backtest tune them. Step 4A.2 requires per-cell, per-night conditions, a score, the top factor named, and documented weights [BP 4A.2]. ADR-0001 fixes that the score is a formula and not a model.

Decision (approved)

The score is the formula below. It is kept simple on purpose so each number can be explained to a visitor.

Status: APPROVED by John on Oct 9, 2026, as written. The weights are STARTING VALUES. They are not validated. The 4B.3 backtest tunes them later, and any change gets a new score_version and an ADR. John approved this specification on Oct 9, 2026. Step 4A.2 may be built; the other Phase A gates still apply.

Ingredients (the four inputs of step 4A.1, nothing else):

1. Cloud cover: NOAA GFS total cloud cover, in percent.
2. The Moon: altitude and illumination from Skyfield with a JPL ephemeris.
3. Light pollution: the VIIRS annual nighttime lights value of the cell.
4. Shower activity: the IMO calendar (shower name, active dates, peak date, rate).

The night: local night at the cell, from sunset to sunrise in the cell's time zone (decided by John, Oct 9, 2026). The night is sampled once an hour in UTC from sunset to sunrise (the hourly sampling is a proposal; it matches the hourly GFS output). `night_date` is the local calendar date on which the sunset falls (proposed).

Step 1. Four factor values, each from 0 (worst) to 1 (best for seeing shooting stars):

Factor value	Definition
`clear_sky`	1 minus (`cloud_cover_pct` / 100), where `cloud_cover_pct` is the mean GFS total cloud cover over the forecast hours that fall inside the night
`moon_dark`	1 minus the mean, over the hourly samples, of `moon_glare_h`. `moon_glare_h` is the Moon illumination in percent divided by 100 when the Moon is above the horizon in that hour, and 0 when it is below
`dark_sky`	1 minus `light_rank`. `light_rank` is the share of cells whose VIIRS light value is lower than this cell's value, among all cells loaded in the run (0 is the darkest cell, 1 the brightest). A rank is used because the VIIRS unit is not yet known
`shower_activity`	0 when no shower is active on `night_date`. Otherwise, for each active shower, multiply (`rate` divided by the largest `rate` in the loaded calendar) by `peak_proximity`, and take the largest result. `peak_proximity` is 1 minus (the days from `night_date` to the peak date, divided by the days from the peak date to the nearer end of the active period on the same side as `night_date`), kept between 0 and 1

Step 2. Combine. Clouds block everything, so `clear_sky` multiplies; the other three are a weighted sum:

`inner = 0.40 x moon_dark + 0.40 x dark_sky + 0.20 x shower_activity` (STARTING WEIGHTS; they add to 1)

`score = 100 x clear_sky x inner` (a number from 0 to 100)

Step 3. Points lost per factor. "Points lost" means how many of the 100 possible points a factor takes away from the score. The four values add up to 100 minus the score:

- `lost_cloud_cover` = 100 x (1 - `clear_sky`)
- `lost_moon` = 100 x `clear_sky` x 0.40 x (1 - `moon_dark`)
- `lost_light_pollution` = 100 x `clear_sky` x 0.40 x (1 - `dark_sky`)
- `lost_shower` = 100 x `clear_sky` x 0.20 x (1 - `shower_activity`)

Step 4. The top factor is the factor with the most points lost. Ties are broken in this order: cloud_cover, moon, light_pollution, no_active_shower. The visitor sees the factor called "no active shower" (stored as `no_active_shower`), because a plain "shower" would read as if the shower were the problem. It is also the top factor when a shower is active but weak. The name and this proposal are part of OQ-02 and wait for John's approval.

Explainability: in Phase B, each explanation phrase quotes values from the list above (percent cloud cover, percent Moon illumination, light rank, shower name and activity, points lost). The score table stores all of them.

What is not decided here: the cloud-to-cell assignment and score grain (OQ-06), the time zone source and sunrise and sunset method (OQ-03), the VIIRS unit (OQ-07), and whether the IMO calendar provides active dates (OQ-14). The radiant position of a shower is not used by this formula and is not stored.

Alternatives considered

1. Cloud cover as one more weighted term in the sum. Not proposed: a fully clouded sky hides every meteor whatever the other factors are, so cloud is a multiplier.
2. An absolute light pollution scale from the VIIRS value. Not proposed yet: the VIIRS unit and range are not known until the file is read (OQ-07). A rank needs no unit. The choice can change after the first-run download.
3. Other weights. The starting values (0.40, 0.40, 0.20) are a proposal with no measured basis. They exist so the pipeline has a documented starting point. The sources record no other candidate weights.
4. A trained model. Rejected by ADR-0001.

Consequences

- If approved, `ScoreConfig` holds these weights with `score_version` set for this version, and the score engine implements Steps 1 to 4.
- If not approved, step 4A.2 does not start.

- The weights are not validated. The Phase B backtest (4B.3) compares scores with known shower peaks and the fireball baseline and may change the weights; a change needs a new score_version and an ADR.
- The "clearly different" acceptance margin (F-A2) is set after this ADR is approved (decided by John, Oct 9, 2026); until then the check is written as in the plan.
- The shower factor needs active dates and a peak date from the IMO calendar (ADR-0005, OQ-14).

`Shooting Star Forecast/adrs/0014-source-licences-and-credit-wording.md`

0014. Source licences and credit wording are checked against each provider's own page

- Status: Accepted. Confirmed by John, Oct 9, 2026. Previously recorded as: Accepted (checked again at John's request, Oct 9, 2026). The JPL item stays UNVERIFIED.
- Project: 4, Shooting star forecast
- Related: ADR-0005, ADR-0012, OQ-18, OQ-23

Context

The requirements and contracts said VIIRS is "public domain" [BP 4A.1] and that JPL and Skyfield wording was not recorded [OQ-18]. Before writing credit lines, each source's own licence or policy page was read. Pages read on Oct 9, 2026:

Source	Page read	What the page says
NOAA GFS on AWS	registry.opendata.aws/noaa-gfs-bdp-pds	"NOAA data disseminated through NODD are open to the public and can be used as desired." "NOAA requests attribution for the use or dissemination of unaltered NOAA data." Do not state or imply endorsement by NOAA. Modified data may not be presented as unaltered. No AWS account needed.
VIIRS nighttime lights (EOG)	eogdata.mines.edu/products/vnl/	"Many of the VIIRS Nighttime Lights data are available under Creative Commons Attribution 4.0 International license." "When using the data please credit the product generation to the Earth Observation Group." The page does not say which files carry this licence and has no statement for 2024 or 2025 annual products.
VIIRS account (EOG)	eogdata.mines.edu/products/register/	"Effective June 1 2026 EOG is limiting programmatic access via OpenID Client to paid subscribers." Footer: copyright 2021 only.
IMO shower calendar	imo.net/members/imo_live_shower_calendar	Copyright 2026 International Meteor Organization. No terms of use or reuse statement.
JPL DE440 ephemeris	ssd.jpl.nasa.gov/doc/de440_de441.html	No copyright or licence statement. Cite Park et al. 2021, The Astronomical Journal 161:105, doi 10.3847/1538-3881/abd414.
Skyfield	github.com/skyfielders/python-skyfield, LICENSE	MIT licence. Copyright 2013 to 2018 Brandon Rhodes.

Not verified: the "public domain" statement for JPL ephemerides. A 2022 JPL email reported in a secondary search summary calls them public domain. No JPL policy page confirmed it, so it is UNVERIFIED. Older ASCII ephemeris file headers carry a Caltech copyright line (1991 to 2005), which is also unresolved.

Decision

1. Each credit line uses the provider's own wording or citation. Wording is not invented.
2. The VIIRS terms note says that many VIIRS nighttime lights products are CC BY 4.0 and that the annual composite's licence is UNVERIFIED. "Public domain" is removed from the requirements, design doc, contracts and ADR-0012.
3. The JPL terms note says UNVERIFIED and gives the citation. The page credits the citation, not a licence.
4. Anything that cannot be checked is marked unverified in the document. Wording is not guessed.

Draft credit lines (for source_credits; confirmed by John, Oct 9, 2026):

- gfs_cloud_cover: "Cloud cover: NOAA Global Forecast System (GFS), via https://registry.opendata.aws/noaa-gfs-bdp-pds. Not endorsed by NOAA."
- viirs_nightlights: "Nighttime lights: VIIRS annual composite, product generated by the Earth Observation Group (EOG)."
- imo_shower_calendar: "Meteor shower calendar: International Meteor Organization (IMO). Factual fields only."
- jpl_ephemeris: "Moon positions: JPL DE440 planetary ephemeris (Park et al., 2021, The Astronomical Journal 161:105, doi:10.3847/1538-3881/abd414), computed with Skyfield (MIT licence)."

Draft terms notes:

- gfs_cloud_cover: NOAA open data; attribution requested for unaltered data; no implied endorsement. Verified on the registry page.
- viirs_nightlights: EOG says many VIIRS nighttime lights products are CC BY 4.0. Annual composite licence UNVERIFIED. Programmatic access paid-only from June 1, 2026 (OQ-23).
- imo_shower_calendar: Copyright 2026 IMO; no reuse terms; store factual fields only.
- jpl_ephemeris: UNVERIFIED. No licence on the JPL SSD page; a 2022 JPL email reported as public domain is not confirmed. Skyfield: MIT.

Alternatives considered

- Keep "public domain" for VIIRS, as the build plan says. Rejected: the provider's own page does not say that.
- Paraphrase the credit lines. Rejected: NOAA and EOG each asked for specific credit wording.

Consequences

- OQ-18 is partly answered by this default. It stays open until John confirms the wording and the unverified items.
- A "public domain" claim is removed from requirements.md, design-doc.md, the VIIRS and source_credits contracts, and ADR-0012.
- OQ-23 records the paid-only programmatic access. It is answered by ADR-0018 (manual download).

Second check (Oct 9, 2026)

- EOG's VNL page was read again and gives the CC BY 4.0 statement for "many" products only. It names no licence for the annual composite. The CC BY 4.0 wording stays, and the annual composite is still unverified.
- A Google Earth Engine catalog page says the V2.2 annual series is public domain. It is a secondary source that conflicts with EOG's own page, so it is not used for the terms note.
- JPL: no licence or policy page could be found. A search returned only the 2022 email and a citation request. The JPL item stays UNVERIFIED, and the citation stays as the credit.

Part 7: Data contracts

Shooting Star Forecast/contracts/feedback_votes.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# ODCS v3. Published table contract (DynamoDB visitor data), Phase B.
# Draft. Privacy follows Project 1 (decided by John, Oct 9, 2026, ADR-0009): anonymous phone ID, limits per visitor and
per day, no internet addresses.
# Limit numbers and other details are open (OQ-15).
apiVersion: v3.0.2
kind: DataContract
id: p4-feedback-votes
name: feedback_votes
version: 0.1.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    Visitors' answers to "did you see one?" (BP 4B.3). Used to count votes and to support the
    backtest note.
  limitations: >
    Votes carry no visitor ID: the random phone ID is used only for limits and sits in the
    separate counters table. No internet addresses are stored. The H3 cell is computed on the phone
    so exact coordinates never reach the server. Limit numbers, the H3 resolution of a vote,
    retention, the lake copy and whether point-in-time recovery is on for this table are open (OQ-15).
  usage: Written by the vote endpoint (openapi.yaml, POST /votes); counted by the backtest and the page.
servers:
  - server: dynamodb-votes
    type: dynamodb
    environment: prod
    description: DynamoDB table, provisioned within the free tier (ADR-0009). Dev and live tables are separate. Point-
in-time recovery is on by platform default; Project 1 turns it off for its hiker reports, and whether Project 4 does is
OQ-15.
schema:
  - name: feedback_votes
    physicalType: table
    description: One item per vote.
    properties:
      - name: vote_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Unique id of the vote. Generation method is a proposal.
      - name: night_date
        logicalType: date
        required: true
        description: The night the visitor answered about. Date convention as in sky_score_by_cell_night (OQ-03).
      - name: cell_h3
        logicalType: string
        required: true
        description: >
          Place of the vote as an H3 cell, computed on the phone. The resolution is OQ-15.
      - name: saw_one
        logicalType: boolean
        required: true
        description: Answer to "did you see one?".
      - name: created_at_utc
        logicalType: timestamp
        required: true
        description: Time the vote was stored, in UTC. Rounding, if any, is OQ-15.
      - name: expires_at_utc
        logicalType: timestamp
        required: false
        description: Time-to-live for automatic deletion. Whether and when raw votes expire is OQ-15.
  - name: feedback_vote_limits
    physicalType: table
    description: >
      Limit counters, stored apart from the votes and expiring (as in Project 1 step 1B.8), so a vote
      never carries an ID that links it to a visitor. Windows and numbers are OQ-15.
    properties:
      - name: visitor_id
        logicalType: string
```

```
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Random ID created on the phone. Never an internet address.
      - name: window
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Counting window (for example a day). Windows and limits are OQ-15.
      - name: count
        logicalType: integer
        required: true
        description: Votes counted in the window.
      - name: expires_at_utc
        logicalType: timestamp
        required: true
        description: Time the counter is deleted. Project 1 expires its counters after 24 hours; the value here is OQ-
15.
team:
  - username: jwasikye
    role: owner
    description: John, owner of the project.
customProperties:
  - property: retention
    value: undecided, see OQ-15
  - property: personalData
    value: none intended; confirm in the security and privacy document (written while building) once OQ-15 is answered.
  - property: openQuestions
    value: [OQ-03, OQ-15]
```

Shooting Star Forecast/contracts/gfs_cloud_cover.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# Open Data Contract Standard (ODCS) v3. Source contract.
# Draft: field names are proposed by the design doc; items marked OQ-nn are open questions in design-doc.md section 16.
apiVersion: v3.0.2
kind: DataContract
id: p4-gfs-cloud-cover
name: gfs_cloud_cover
version: 0.1.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    NOAA GFS total cloud cover for the entire atmosphere, in percent, hourly to 120 hours,
    about 28 km (0.25 degree) cells. Input to the sky score (design-doc.md section 9).
  limitations: >
    Cells are about 28 km wide, so cloud cover is the same for every finer cell inside one
    H3 resolution 4 cell (ADR-0004). File format, file naming and reader are confirmed by the
    Phase A first-run check (OQ-04). NOAA asks for attribution.
  usage: >
    Read by the nightly pipeline only. Not served to visitors directly.
servers:
  - server: noaa-gfs-open-data
    type: s3
    location: s3://noaa-gfs-bdp-pds/
    environment: prod
    description: AWS Open Data bucket noaa-gfs-bdp-pds in us-east-1. No key needed to read (open data); no key is
recorded in the sources.
schema:
  - name: gfs_cloud_cover_raw_manifest
    physicalType: table
    description: >
      One row per downloaded source file (engineering standard 1: URL, fetch time, hash).
      Raw files are immutable.
    properties:
      - name: source_url
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Full S3 URL of the downloaded file.
      - name: model_run_time_utc
```

```
        logicalType: timestamp
        required: true
        description: GFS model run (one of four a day) the file belongs to, in UTC.
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
        description: Time the file was downloaded, in UTC.
      - name: content_hash
        logicalType: string
        required: true
        description: Hash of the file contents (engineering standard 1). The hash algorithm is not recorded in the
sources and is undecided (platform decision).
      - name: size_bytes
        logicalType: integer
        required: true
        description: File size. The size of a GFS run's cloud file is not published in the sources (OQ-05).
  - name: gfs_cloud_cover_h3
    physicalType: table
    description: >
      Cloud cover per H3 resolution 4 cell, per valid time, per model run. Written to Iceberg
      (Glue catalog). A run's partition is overwritten on rerun (engineering standard 3).
    properties:
      - name: cell_h3_r4
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: H3 index at resolution 4.
      - name: valid_time_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Time the forecast is valid for, in UTC (event time, standard 2).
      - name: model_run_time_utc
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: GFS run that produced the value, in UTC.
      - name: total_cloud_cover_pct
        logicalType: number
        required: true
        description: Total cloud cover for the entire atmosphere, in percent.
        logicalTypeOptions:
          minimum: 0
          maximum: 100
        quality:
          - type: library
            rule: nullValues
            mustBe: 0
      - name: source_key
        logicalType: string
        required: true
        description: Source URL of the file the value came from (engineering standard 2).
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
        description: Load time (F-A1 acceptance criterion).
      - name: forecast_hour
        logicalType: integer
        required: true
        description: Hours after the model run, 0 to 120 (hourly to 120 hours is a confirmed fact).
        logicalTypeOptions:
          minimum: 0
          maximum: 120
    quality:
      - type: library
        rule: duplicateValues
        mustBe: 0
        description: Unique on (cell_h3_r4, valid_time_utc, model_run_time_utc).
slaProperties:
  - property: frequency
    value: four model runs a day (source side)
    description: GFS publishes four runs a day. The pipeline reads once a night (ADR-0007).
team:
```

```
  - username: jwasikye
    role: owner
    description: John, owner of the project.
support:
  - channel: github-issues
    tool: github
    description: One de-duplicated issue per failed run or missed freshness target (platform step 1B.9).
customProperties:
  - property: freshnessTarget
    value: undecided, see OQ-20
  - property: sourceTerms
    value: NOAA asks for attribution (DP Cloud cover data). Exact credit line wording is OQ-18.
  - property: h3Resolution
    value: 4
  - property: openQuestions
    value: [OQ-04, OQ-05, OQ-18, OQ-20]
```

Shooting Star Forecast/contracts/imo_shower_calendar.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# ODCS v3. Source contract.
# Draft: only the fields the score needs, each with its source (decided by John, Oct 9, 2026, ADR-0005).
apiVersion: v3.0.2
kind: DataContract
id: p4-imo-shower-calendar
name: imo_shower_calendar
version: 0.1.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    International Meteor Organization (IMO) shower calendar, kept as a small reference table of
    the factual fields the score needs (shower name, active dates, peak date, rate). Input to the shower term of the
sky score.
  limitations: >
    The IMO site has a copyright notice and no reuse terms, so only factual fields are stored and
    the IMO is credited (ADR-0005). Radiant position and other descriptive fields are not stored (the proposed formula
does not use them).
    Whether the IMO calendar provides active dates is OQ-14. The meaning of the published rate figure is confirmed when
the table is built (OQ-14).
    How the calendar is retrieved from the IMO site (format, URL) is not recorded (OQ-14).
  usage: Read by the nightly pipeline.
schema:
  - name: imo_shower_calendar_raw_manifest
    physicalType: table
    description: One row per retrieved page or file (source URL, fetch time, hash), standard 1.
    properties:
      - name: source_url
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: URL the calendar was retrieved from.
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
        description: Retrieval time, in UTC.
      - name: content_hash
        logicalType: string
        required: true
        description: Hash of the retrieved content (engineering standard 1). Hash algorithm undecided (platform
decision).
  - name: imo_shower_calendar
    physicalType: table
    description: Reference table of meteor showers. Small.
    properties:
      - name: shower_name
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Name of the shower as published by the IMO (factual).
      - name: active_start
        logicalType: date
```

```
        required: true
        description: First date the shower is active, as published by the IMO. Used by the shower factor (ADR-0010).
Availability in the IMO calendar is OQ-14.
      - name: active_end
        logicalType: date
        required: true
        description: Last date the shower is active, as published by the IMO.
      - name: peak_date
        logicalType: date
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Peak date for a calendar year (factual). Stored as a date in UTC terms; the calendar's own time
convention is confirmed when built (OQ-14).
      - name: rate
        logicalType: number
        required: false
        description: Rate published for the shower (factual). Meaning (for example hourly rate under ideal conditions)
is confirmed when built (OQ-14).
        quality:
          - type: library
            rule: invalidValues
            mustBe: 0
            arguments:
              minValue: 0
            description: A rate cannot be negative.
      - name: source_key
        logicalType: string
        required: true
        description: Source URL of the IMO calendar the row came from; the source of every field in the row
(engineering standard 2).
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
        description: Load time (F-A1 acceptance criterion).
    quality:
      - type: library
        rule: duplicateValues
        mustBe: 0
        description: Unique on (shower_name, peak_date).
team:
  - username: jwasikye
    role: owner
    description: John, owner of the project.
customProperties:
  - property: updateFrequency
    value: undecided, see OQ-14
  - property: sourceTerms
    value: Copyright notice, no reuse terms. Store factual fields only and credit the IMO (BP Risks table). Credit line
wording is OQ-18.
  - property: openQuestions
    value: [OQ-14, OQ-18]
```

Shooting Star Forecast/contracts/jpl_ephemeris.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# ODCS v3. Source contract.
# Draft: the ephemeris file is not yet chosen (OQ-08).
apiVersion: v3.0.2
kind: DataContract
id: p4-jpl-ephemeris
name: jpl_ephemeris
version: 0.1.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    JPL ephemeris file read by Skyfield to compute Moon altitude and illumination (ADR-0003).
  limitations: >
    Which ephemeris file is used, its size, its download location and its terms are not recorded
    in the sources (OQ-08). The Moon values computed from it are not a source; they are an
    intermediate computed table (design-doc.md section 9).
  usage: Read by the nightly pipeline when Moon values are computed.
schema:
```

```
  - name: jpl_ephemeris_manifest
    physicalType: table
    description: >
      One row per ephemeris file used (file name, source URL, fetch time, hash). The file is a
      static archive, so a manifest is kept in line with standard 1; whether the file itself is also
      stored is OQ-08.
    properties:
      - name: file_name
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Name of the ephemeris file.
      - name: source_url
        logicalType: string
        required: true
        description: URL the file was downloaded from.
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
        description: Download time, in UTC.
      - name: content_hash
        logicalType: string
        required: true
        description: Hash of the file contents (engineering standard 1). Hash algorithm undecided (platform decision).
      - name: size_bytes
        logicalType: integer
        required: true
        description: File size.
slaProperties:
  - property: frequency
    value: not applicable
    description: Static reference file; changes only if the file choice changes (new ADR).
team:
  - username: jwasikye
    role: owner
    description: John, owner of the project.
customProperties:
  - property: sourceTerms
    value: JPL SSD page has no licence statement; cite Park et al. 2021. The public-domain claim is unverified.
Skyfield is MIT (ADR-0014). Credit wording is OQ-18.
  - property: accuracyFact
    value: Skyfield with JPL ephemerides agrees with the U.S. Naval Observatory to within 0.00001 arcseconds (DP
Answered by research).
  - property: openQuestions
    value: [OQ-08, OQ-18]
```

Shooting Star Forecast/contracts/sky_score_by_cell_night.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# ODCS v3. Published table contract.
# Draft. The columns follow the score formula in ADR-0010 (approved by John, Oct 9, 2026).
# The grain (H3 resolution) is an open question (OQ-06).
apiVersion: v3.0.2
kind: DataContract
id: p4-sky-score-by-cell-night
name: sky_score_by_cell_night
version: 0.2.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    Per-cell, per-night conditions, sky score and top factor. Written by Spark to Iceberg, then
    staged, tested and built by dbt as a mart and published as Parquet by the nightly run. Read
    by Lambda with DuckDB to answer lookups with the session box off.
  limitations: >
    The score is the output of the formula in ADR-0010 (approved Oct 9, 2026), whose weights are
    STARTING VALUES that are not validated and may be tuned by the 4B.3 backtest. The H3
    resolution of the grain is open (OQ-06). Cloud cover is shared by all cells inside one H3
    resolution 4 parent (ADR-0004). "Night" is the local night at the place, sunset to sunrise
    (ADR-0013). Earlier nights stay in the published data so a late lookup can be served the
    latest earlier row (ADR-0011).
  usage: >
    Served by the forecast endpoint (openapi.yaml, GET /forecast). Published only after the
```

```
    Great Expectations publish gate passes (engineering standard 4).
schema:
  - name: sky_score_by_cell_night
    physicalType: parquet
    description: One row per cell per night. A night's partition is overwritten on rerun (engineering standard 3).
    properties:
      - name: cell_h3
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: H3 index at the score grain. Resolution is OQ-06.
      - name: night_date
        logicalType: date
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Local calendar date on which the night's sunset falls at the cell (proposed convention, ADR-0013).
      - name: night_start_utc
        logicalType: timestamp
        required: true
        description: Sunset at the cell, in UTC.
      - name: night_end_utc
        logicalType: timestamp
        required: true
        description: Following sunrise at the cell, in UTC.
      - name: forecast_run_time_utc
        logicalType: timestamp
        required: true
        description: GFS model run used for the cloud cover, in UTC.
      - name: cloud_cover_pct
        logicalType: number
        required: true
        description: Mean GFS total cloud cover over the forecast hours inside the night, from gfs_cloud_cover via
cell_to_parent.
        logicalTypeOptions:
          minimum: 0
          maximum: 100
        quality:
          - type: library
            rule: nullValues
            mustBe: 0
      - name: moon_illumination_pct
        logicalType: number
        required: true
        description: Mean Moon illuminated fraction in percent over the night samples, from Skyfield (ADR-0003).
        logicalTypeOptions:
          minimum: 0
          maximum: 100
      - name: moon_altitude_max_deg
        logicalType: number
        required: true
        description: Highest Moon altitude in degrees over the night samples.
        logicalTypeOptions:
          minimum: -90
          maximum: 90
      - name: moon_glare
        logicalType: number
        required: true
        description: Mean over the night samples of (Moon illumination / 100 when the Moon is above the horizon, else
0). Equals 1 minus moon_dark in ADR-0010.
        logicalTypeOptions:
          minimum: 0
          maximum: 1
      - name: light_value
        logicalType: number
        required: true
        description: Light pollution value from viirs_nightlights at resolution 8. Unit is OQ-07.
      - name: light_rank
        logicalType: number
        required: true
        description: Percentile rank (0 to 1) of light_value among all cells loaded in the run. Equals 1 minus dark_sky
in ADR-0010.
        logicalTypeOptions:
          minimum: 0
          maximum: 1
```

```
      - name: shower_name
        logicalType: string
        required: false
        description: Strongest active shower that night (largest rate x peak proximity), if any.
      - name: shower_activity
        logicalType: number
        required: true
        description: Shower activity from 0 to 1 per ADR-0010 (0 when no shower is active).
        logicalTypeOptions:
          minimum: 0
          maximum: 1
      - name: score
        logicalType: number
        required: true
        description: The sky score, 0 to 100 (proposed scale, ADR-0010).
        logicalTypeOptions:
          minimum: 0
          maximum: 100
      - name: top_factor
        logicalType: string
        required: true
        description: The factor with the most points lost; ties in the order cloud_cover, moon, light_pollution,
no_active_shower.
        logicalTypeOptions:
          pattern: "^(cloud_cover|moon|light_pollution|no_active_shower)$"
        quality:
          - type: library
            rule: nullValues
            mustBe: 0
      - name: lost_cloud_cover
        logicalType: number
        required: true
        description: Points lost to cloud cover (ADR-0010). The four lost_ columns add up to 100 minus score.
        logicalTypeOptions:
          minimum: 0
          maximum: 100
      - name: lost_moon
        logicalType: number
        required: true
        description: Points lost to the Moon.
        logicalTypeOptions:
          minimum: 0
          maximum: 100
      - name: lost_light_pollution
        logicalType: number
        required: true
        description: Points lost to light pollution.
        logicalTypeOptions:
          minimum: 0
          maximum: 100
      - name: lost_shower
        logicalType: number
        required: true
        description: Points lost because no strong shower is active.
        logicalTypeOptions:
          minimum: 0
          maximum: 100
      - name: score_version
        logicalType: string
        required: true
        description: Version of the formula and weights that produced the row.
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
        description: Time the row was produced, in UTC. The page shows it as the time of the last good run.
    quality:
      - type: library
        rule: duplicateValues
        mustBe: 0
        description: Unique on (cell_h3, night_date) (dbt unique and not_null on keys, standard 6).
      - type: library
        rule: rowCount
        mustBeGreaterThan: 0
        description: An empty result must not publish.
slaProperties:
  - property: frequency
```

```
    value: 1
    unit: d
    description: Produced by the nightly run at 10:00 UTC (ADR-0007).
team:
  - username: jwasikye
    role: owner
    description: John, owner of the project.
customProperties:
  - property: freshnessTarget
    value: undecided, see OQ-20
  - property: topFactorValues
    value: cloud_cover, moon, light_pollution, no_active_shower (ADR-0010, approved)
  - property: openQuestions
    value: [OQ-02, OQ-03, OQ-06, OQ-07, OQ-14, OQ-20]
```

Shooting Star Forecast/contracts/source_credits.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# ODCS v3. Published table contract.
# Draft: wording of credit lines and terms notes is OQ-18.
apiVersion: v3.0.2
kind: DataContract
id: p4-source-credits
name: source_credits
version: 0.1.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    One row per input with its terms note and credit line, so the page can credit every source
    from the data it serves (BP 4A.1: each input has a terms note and credit line; BP 4A.3:
    every source credited).
  limitations: The wording of each credit line and terms note is not yet written (OQ-18).
  usage: Read by the forecast endpoint and returned with every answer.
schema:
  - name: source_credits
    physicalType: parquet
    description: Small reference table, four rows in Phase A.
    properties:
      - name: source_id
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: One of gfs_cloud_cover, viirs_nightlights, imo_shower_calendar, jpl_ephemeris.
        logicalTypeOptions:
          pattern: "^(gfs_cloud_cover|viirs_nightlights|imo_shower_calendar|jpl_ephemeris)$"
      - name: display_name
        logicalType: string
        required: true
        description: Name shown on the page.
      - name: credit_line
        logicalType: string
        required: true
        description: Credit text shown on the page.
        quality:
          - type: library
            rule: missingValues
            mustBe: 0
            arguments:
              missingValues: [""]
      - name: terms_note
        logicalType: string
        required: true
        description: Terms note for the source (for example NOAA asks for attribution; VIIRS CC BY 4.0 for many
products, annual composite unverified; IMO factual fields only).
        quality:
          - type: library
            rule: missingValues
            mustBe: 0
            arguments:
              missingValues: [""]
      - name: source_url
        logicalType: string
```

```
        required: false
        description: Link to the source shown with the credit.
    quality:
      - type: library
        rule: rowCount
        mustBe: 4
        description: One row per Phase A input.
slaProperties:
  - property: frequency
    value: on change
    description: Changes only when a source or its terms change.
team:
  - username: jwasikye
    role: owner
    description: John, owner of the project.
customProperties:
  - property: openQuestions
    value: [OQ-18]
```

Shooting Star Forecast/contracts/viirs_nightlights.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
# ODCS v3. Source contract.
# Draft: the VIIRS file has not been downloaded yet. Its format is an open question (OQ-01, OQ-07).
# Decided by John, Oct 9, 2026 (ADR-0012): keep a manifest (URL, size, hash) and re-download once a year.
apiVersion: v3.0.2
kind: DataContract
id: p4-viirs-nightlights
name: viirs_nightlights
version: 0.1.0
status: draft
domain: shooting-star-forecast
dataProduct: project4-sky-score
tenant: portfolio
description:
  purpose: >
    VIIRS annual nighttime lights composite (licence unverified; see ADR-0014), used as the light pollution input
    to the sky score at H3 resolution 8.
  limitations: >
    Latitude coverage is unknown until the file is downloaded (OQ-01). Product version, year,
    variable, unit and file format are not recorded in the sources (OQ-07). A free Earth
    Observation Group account is needed to download it. File size is not published (OQ-05). Programmatic access is
paid-only from June 1, 2026 (OQ-23). No 2024 or 2025 annual product is listed (OQ-24).
  usage: >
    Read by the pipeline once a year, when the file is re-downloaded (ADR-0012). The date of the yearly load depends on
the release of a new composite (OQ-07).
servers:
  - server: eog-download
    type: custom
    environment: prod
    description: Earth Observation Group download behind a free login (programmatic access is paid-only from June 1,
2026; OQ-23). The file is downloaded by hand (ADR-0018), so no token is stored. The schedule is open (OQ-07).
schema:
  - name: viirs_raw_manifest
    physicalType: table
    description: >
      One row per downloaded file: source URL, fetch time, hash, size. Only this manifest is kept,
      not the raw file (decided by John, Oct 9, 2026, ADR-0012).
    properties:
      - name: source_url
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: URL the file was downloaded from.
      - name: fetched_at_utc
        logicalType: timestamp
        required: true
        description: Download time, in UTC.
      - name: content_hash
        logicalType: string
        required: true
        description: Hash of the file contents (engineering standard 1). Hash algorithm undecided (platform decision).
      - name: size_bytes
        logicalType: integer
        required: true
```

```
        description: File size (OQ-05).
  - name: viirs_light_h3
    physicalType: table
    description: Light pollution value per H3 resolution 8 cell for one annual composite. Written to Iceberg (Glue
catalog).
    properties:
      - name: cell_h3_r8
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: H3 index at resolution 8.
      - name: composite_year
        logicalType: integer
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Year of the annual composite. Which year is loaded is OQ-07.
      - name: radiance_value
        logicalType: number
        required: true
        description: >
          Light value of the cell. Variable, unit and the aggregation from raster pixels to the
          cell are not recorded (OQ-07). Valid range is set once the file is read.
        quality:
          - type: library
            rule: nullValues
            mustBe: 0
      - name: source_key
        logicalType: string
        required: true
        description: Source URL of the file the value came from (engineering standard 2).
      - name: loaded_at_utc
        logicalType: timestamp
        required: true
        description: Load time (F-A1 acceptance criterion).
    quality:
      - type: library
        rule: duplicateValues
        mustBe: 0
        description: Unique on (cell_h3_r8, composite_year).
slaProperties:
  - property: frequency
    value: 1
    unit: y
    description: The composite is annual (source side). The file is re-downloaded once a year (ADR-0012); the release
timing is OQ-07.
team:
  - username: jwasikye
    role: owner
    description: John, owner of the project.
customProperties:
  - property: sourceTerms
    value: EOG says many VNL products are CC BY 4.0. The annual composite licence is unverified (ADR-0014). Credit line
wording is OQ-18.
  - property: h3Resolution
    value: 8
  - property: openQuestions
    value: [OQ-01, OQ-05, OQ-07, OQ-18, OQ-23, OQ-24]
```

Part 8: API specification

Shooting Star Forecast/openapi.yaml

```
# Project 4: Shooting star forecast. Scope: this project only. Shared platform documents are in ../Platform/.
openapi: 3.0.3
info:
  title: "Project 4: Shooting star forecast API"
  version: 0.2.0-draft
  description: |
    Public endpoints for Project 4. Draft. Path names, parameters and response shapes are proposed by
    design-doc.md section 10 and must be aligned with the platform serving template
    (../Platform/openapi-serving-template.yaml, not yet written). Open items: OQ-03, OQ-06, OQ-10,
    OQ-15, OQ-19. The score fields follow the formula in ADR-0010 (approved by John, Oct 9, 2026).

    Served by FastAPI on Lambda behind API Gateway (HTTP API type, for public pages) with DuckDB
    over the published Parquet (BP 0.4). Works with the session box off. The first request after a
    period of inactivity may take up to 10 seconds while the service initializes.

    No API keys: the endpoints back a public page. The REST type of API Gateway (keys and usage plans)
    is used by Project 5 only; no key requirement is recorded for Project 4.
servers:
  - url: https://api.placeholder.invalid
    description: Placeholder. The real URL is set at deploy by the serving template. Dev and live are separate
deployments.
tags:
  - name: forecast
    description: Phase A.
  - name: votes
    description: Phase B.
paths:
  /forecast:
    get:
      tags: [forecast]
      operationId: getForecast
      summary: Tonight's sky score and top factor for a place
      description: |
        Returns the score and the top factor for the cell that contains the given point, with every
        source credited. Phase A has data for the lower 48 states (Alaska and Hawaii follow the VIIRS
        latitude check); Phase B has data for every land cell the VIIRS data covers (ADR-0006).
        "Tonight" is the local night at the place, sunset to sunrise (ADR-0013).

        Late data (ADR-0011, decided by John, Oct 9, 2026): when no row exists for tonight, the most
        recent earlier row is returned with late set to true and last_good_run_time_utc, and the page
        shows a visible late notice. Data is never hidden.

        How the point becomes a cell and the score grain are open (OQ-06, OQ-10).
      parameters:
        - name: lat
          in: query
          required: true
          description: Latitude in degrees. Whether the page sends coordinates, a place string or a cell is OQ-10.
          schema:
            type: number
            format: double
            minimum: -90
            maximum: 90
        - name: lon
          in: query
          required: true
          description: Longitude in degrees.
          schema:
            type: number
            format: double
            minimum: -180
            maximum: 180
      responses:
        "200":
          description: A row exists for the point, either for tonight or, with late set, for an earlier night.
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/ForecastResponse"
        "400":
          description: Missing or out-of-range lat or lon.
          content:
            application/json:
```

```
              schema:
                $ref: "#/components/schemas/Error"
        "404":
          description: |
            No published data for the cell at all: the point is outside the covered area (Phase A:
            outside the lower 48 until Alaska and Hawaii are added; Phase B: beyond the VIIRS latitude
            coverage), or the cell has never had a row. A missing tonight alone does not return 404.
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/NoData"
        "429":
          description: Throttled by API Gateway. Throttle values are not recorded (OQ-19).
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/Error"
        "500":
          description: Unexpected error.
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/Error"
  /votes:
    post:
      tags: [votes]
      operationId: postVote
      summary: Store a "did you see one?" answer (Phase B)
      x-phase: B
      description: |
        Stores one vote in DynamoDB (ADR-0009). Privacy follows Project 1 (decided by John, Oct 9,
        2026): an anonymous random phone ID used only for limits, limits per visitor and per day, no
        internet addresses. The cell is computed on the phone. The visitor_id is checked against the
        separate counters table and is not stored with the vote. Limit numbers and other details are
        open (OQ-15). Do not build the endpoint until OQ-15 is answered.
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/VoteRequest"
      responses:
        "202":
          description: The vote was accepted and stored.
        "400":
          description: Invalid body.
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/Error"
        "429":
          description: Vote limit reached or API Gateway throttle. Limits are not recorded (OQ-15, OQ-19).
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/Error"
        "500":
          description: Unexpected error.
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/Error"
components:
  schemas:
    ForecastResponse:
      type: object
      required: [night_date, late, last_good_run_time_utc, score, top_factor, conditions, points_lost, credits,
forecast_run_time_utc]
      properties:
        night_date:
          type: string
          format: date
          description: The night the served score is for (local date of the sunset at the place).
        late:
          type: boolean
          description: True when night_date is earlier than tonight's night_date at the place. The page then shows a
visible late notice.
        last_good_run_time_utc:
```

```
          type: string
          format: date-time
          description: Time of the last good run that produced the served row (loaded_at_utc). Shown on the late notice.
UTC; the page converts to local time.
        score:
          type: number
          minimum: 0
          maximum: 100
          description: The sky score, 0 to 100 (proposed scale, ADR-0010).
        top_factor:
          $ref: "#/components/schemas/Factor"
        conditions:
          $ref: "#/components/schemas/Conditions"
        points_lost:
          $ref: "#/components/schemas/PointsLost"
        forecast_run_time_utc:
          type: string
          format: date-time
          description: GFS run the cloud cover came from.
        night_start_utc:
          type: string
          format: date-time
          description: Sunset at the place, in UTC.
        night_end_utc:
          type: string
          format: date-time
          description: Following sunrise at the place, in UTC.
        explanation:
          type: string
          description: Pre-written phrase filled with computed values (ADR-0002). Phase B; absent in Phase A.
          x-phase: B
        coverage_note:
          type: string
          description: Notice about the data's latitude limits when the point is near them. Phase B; absent in Phase A.
          x-phase: B
        credits:
          type: array
          description: One entry per source, from the source_credits table. Every source is credited.
          items:
            $ref: "#/components/schemas/Credit"
    Conditions:
      type: object
      description: The inputs behind the score, as stored in sky_score_by_cell_night.
      required: [cloud_cover_pct, moon_illumination_pct, moon_altitude_max_deg, moon_glare, light_value, light_rank,
shower_activity]
      properties:
        cloud_cover_pct:
          type: number
          minimum: 0
          maximum: 100
        moon_illumination_pct:
          type: number
          minimum: 0
          maximum: 100
        moon_altitude_max_deg:
          type: number
          minimum: -90
          maximum: 90
        moon_glare:
          type: number
          minimum: 0
          maximum: 1
        light_value:
          type: number
          description: VIIRS light value. Unit is OQ-07.
        light_rank:
          type: number
          minimum: 0
          maximum: 1
        shower_name:
          type: string
          nullable: true
        shower_activity:
          type: number
          minimum: 0
          maximum: 1
    PointsLost:
      type: object
```

```
      description: Points lost per factor; the four values add up to 100 minus score (ADR-0010).
      required: [cloud_cover, moon, light_pollution, no_active_shower]
      properties:
        cloud_cover:
          type: number
        moon:
          type: number
        light_pollution:
          type: number
        no_active_shower:
          type: number
    Factor:
      type: string
      description: The factor with the most points lost (ADR-0010, approved). The visitor sees no_active_shower as "no
active shower"; the name was approved with the formula (OQ-02).
      enum: [cloud_cover, moon, light_pollution, no_active_shower]
    Credit:
      type: object
      required: [source_id, display_name, credit_line, terms_note]
      properties:
        source_id:
          type: string
          enum: [gfs_cloud_cover, viirs_nightlights, imo_shower_calendar, jpl_ephemeris]
        display_name:
          type: string
        credit_line:
          type: string
        terms_note:
          type: string
        source_url:
          type: string
          nullable: true
    NoData:
      type: object
      required: [reason, message]
      properties:
        reason:
          type: string
          enum: [outside_coverage, no_data_for_cell]
        message:
          type: string
          description: Text for the visitor. Wording is not decided.
        coverage_note:
          type: string
          description: Latitude limit notice, when the reason is outside_coverage (Phase B).
    VoteRequest:
      type: object
      description: Provisional (OQ-15).
      required: [night_date, cell_h3, saw_one, visitor_id]
      properties:
        night_date:
          type: string
          format: date
        cell_h3:
          type: string
          description: H3 cell of the place, computed on the phone. Resolution is OQ-15.
        saw_one:
          type: boolean
        visitor_id:
          type: string
          description: Random ID created on the phone, used only for limits and not stored with the vote. Never an
internet address.
    Error:
      type: object
      required: [message]
      properties:
        message:
          type: string
        code:
          type: string
```

Part 9: Changelog

Shooting Star Forecast/changelog.md

Changelog: Project 4, Shooting star forecast

Oct 9, 2026: defaults confirmed

- John confirmed ADR-0014 (licences and draft credit lines) and ADR-0017 (what "tonight" means). Their status is now Accepted.

Rev 8, Oct 9, 2026 (operations and security documents added)

- Added `operations.md` and `security-and-privacy.md`. The documentation plan requires both for every project, and Project 4 did not have them. Every value comes from the design doc, the ADRs and the contracts. Values that are not decided name their open question.
- `full-documentation.pdf` is rebuilt from this folder for rev 8. It includes the two new documents.
- The PDFs are now one file in this folder, `full-documentation.pdf`, as in the other projects. The `pdfs/` folder is removed. The original PDF and revisions 2 to 6 were removed as old versions. Rev 7 was the last of them and is replaced by this rebuild.
- README.md lists the new documents and describes the PDF.
- No change to decisions, counts or open questions.

Rev 7, Oct 9, 2026 (platform decisions approved)

- John approved OQ-P01 (SNS email and the metrics table in the lake; ADR-0015) and OQ-P02 in part (the session limits; the idle-alarm duration is still to be measured). The platform set's text now records this. The platform set as a whole is still a draft.

Rev 6, Oct 9, 2026 (index corrected)

- Fixed a duplicated line in README.md about the shared platform folder.
- README.md's PDF table now lists rev 5 as the current revision.
- No change to decisions, counts or open questions.

Rev 5, Oct 9, 2026 (decision log split)

- The shared decision logs were split into one `decisions.md` per project. Project 4's file is `decisions.md` in this folder: 10 answered rows and the 6 open items from the first review.
- The original shared logs are kept unchanged in `Project_Documentation\_archive\`.
- README.md now points to `decisions.md` in place of the shared logs.
- No change to decisions, counts or open questions.

Rev 4, Oct 9, 2026 (labels, index, PDFs gathered)

- Every contract and the OpenAPI file carry a "Project 4" label on their first line; the OpenAPI title says Project 4.
- Project 4's three PDFs moved from `Project_Documentation\PDFs\` to `Shooting Star Forecast\pdfs\`, names unchanged. Rev 4 is built there.
- README.md indexes every file in the folder and lists Project 4's rows in the shared decision logs.
- No change to decisions, counts or open questions.

Rev 3, Oct 9, 2026 (platform set drafted; approval-ready)

- The platform document set now exists in ../Platform/ (requirements, 17-section design doc, 17 ADRs, three shared contracts, the serving-template API spec, changelog). The links from this project now resolve.

- Fixed the ADR-0008 links to the platform and to Project 1 (they point one folder too high).
- requirements.md and design-doc.md no longer say the platform documents are missing.
- Added a "Ready for John's approval" checklist to README.md: what to approve, the decisions and spikes that still gate Phase A, and the platform gates.
- Counts are unchanged from rev 2: 24 question IDs; 4 answered in full; 20 open (8 partly answered).
- Rev 2 PDF is kept; rev 3 is the current PDF.

Rev 2 follow-up, Oct 9, 2026 (John's decisions)

Decided

- Score formula approved by John as written (ADR-0010 is Accepted). The weights stay starting values for the 4B.3 backtest to tune. Docs that said "proposed" or "awaiting approval" now say approved. The next open item is the numeric margin (OQ-13).
- OQ-23: free EOG login; the VIIRS file is downloaded by hand. No automated download and no EOG credential in Parameter Store (ADR-0018). Updated design doc sections 7.1, 7.3, 7.17, 8.4, 11, the context diagram label, the VIIRS contract and ADR-0012.
- OQ-22: before sunrise, "tonight" is the night still in progress (ADR-0017, Default; Claude picked it at John's request).
- Superseded duplicate ADR-0006 file deleted at John's request. ADR-0016 updated. Its content matched the lower-48 decision, which is kept.

Checked again

- ADR-0014 (licences and credit wording) accepted. VIIRS: the EOG page still gives CC BY 4.0 only for "many" products; no licence for the annual composite. A Google Earth Engine catalog page says public domain, but it is a secondary source that conflicts with EOG, so it is not used. JPL: no licence or policy page found; stays UNVERIFIED.
- OQ-24 partly answered. Google Earth Engine's catalog lists the V2.2 series through 2024 (secondary source). The EOG page gives 2012 to 2020 for V2 and no years for V2.2. No 2025 product was found. Correction: ADR-0012 earlier said the EOG page lists annual composites only to 2020; that was wrong and is fixed.

Counts (section 16)

- Question IDs: 24 (OQ-01 to OQ-24).
- Answered in full: 4 (OQ-02, OQ-11, OQ-22, OQ-23).
- Open: 20. Of these, 8 have a partial answer (OQ-03, OQ-06, OQ-07, OQ-13, OQ-14, OQ-15, OQ-18, OQ-24). The other 12 have no answer (OQ-01, OQ-04, OQ-05, OQ-08, OQ-09, OQ-10, OQ-12, OQ-16, OQ-17, OQ-19, OQ-20, OQ-21).
- ADR files: 18, all active.

Still open for John

- A numeric margin for the "clearly different" check (OQ-13). The formula is now approved, so this is next.
- Confirm the unverified items in ADR-0014 (the VIIRS annual composite licence and JPL), if you want them closed.
- OQ-24: check the EOG page for the V2.2 years and for any 2025 product before the first VIIRS load.

Rev 2, Oct 9, 2026

Source checks, decisions from John's answers of Oct 9, and fixes from the review before the PDF.

Added

- ADR-0014: source licences and credit wording, checked against each provider's own page. Status: Default, for John to confirm.
- ADR-0015: the "clearly different" margin waits for the approved formula; Phase C stays unspecified (answers 25, 26, 33).
- ADR-0016: the second ADR-0006 file is marked superseded (deleted later the same day, see the follow-up above).

- OQ-11 restored in the section 16 table. It had been missing from the numbering. Answer 1 (ADR-0011) recorded.
- OQ-22: which night is "tonight" between midnight and sunrise.
- OQ-23: EOG limited programmatic VIIRS access to paid subscribers from June 1, 2026.
- OQ-24: no 2024 or 2025 annual VIIRS composite is listed on the EOG VNL page.
- README.md and this changelog.

Changed

- VIIRS is no longer described as public domain. EOG's VNL page says many VIIRS nighttime lights products are CC BY 4.0, and the annual composite's licence is unverified. Changed in requirements.md, design-doc.md, ADR-0012, viirs_nightlights.yaml and source_credits.yaml.
- JPL ephemeris licence is marked UNVERIFIED. The JPL SSD page has no licence statement. Skyfield is MIT.
- Section 16 keeps answered questions in the table, with the answer and ADR number, instead of removing them. The intro sentence says so.
- OQ-02, 03, 06, 07, 13, 14, 15 and 18 now show the answer or default that applies and what is still open.
- ADR-0013 notes that "tonight" before sunrise is open (OQ-22).
- Every place that described the VIIRS download as a free account now points to OQ-23, including the context diagram label.
- ADR-0008 Project 1 link fixed (it pointed inside Shooting Star Forecast/).

Counts (section 16)

- Question IDs: 24 (OQ-01 to OQ-24).
- Answered in full: 1 (OQ-11).
- Open: 23. Of these, 8 have a partial answer or default recorded (OQ-02, 03, 06, 07, 13, 14, 15, 18). The other 15 have no answer.
- Phase C: unspecified by decision (ADR-0015); no OQ row.
- ADR files: 17. Active decisions: 16 (ADR-0006 appears twice; one file is superseded).

Still open for John

- Approve or change ADR-0010 (the formula). The margin (OQ-13) waits on it.
- Confirm ADR-0014 (credit wording and the unverified items).
- Decide OQ-23 (how the VIIRS file is downloaded, given paid-only programmatic access).
- Decide OQ-22 (which night is "tonight" before sunrise).
- Check whether a newer annual VIIRS product exists (OQ-24).

Known gaps, not fixed here

- Platform documents (../Platform/...) do not exist yet. Links to them are placeholders, as the build plan says.
- The changelog for Project 4 started at rev 2. No earlier entries were recorded in the docs set.